

Highlights

- Frames the multi-agent LLM defense pipeline as an object whose own security, under insider compromise and second-order injection, must be analysed.
- **Aegis-Agency:** Byzantine-robust verdict aggregation, payload isolation, and hardened judges, generalised from jailbreak to prompt-injection/task-integrity.
- Decision-integrity guarantee for robust aggregation in a semantic verdict space under explicit honest-concentration and margin assumptions.
- Correlated-failure analysis that formalises the backbone-diversity requirement and shows added judges do not help once errors are correlated.
- Adaptive-attack evaluation protocol and a security-versus-cost frontier over the number of hardened defenders.

Aegis-Agency: Byzantine-Robust, Injection-Hardened Multi-Agent Defense Pipelines for Large Language Models

Author One^{a,*}, Author Two^a, Author Three^b

^a*Department of Computing, University Placeholder, City, Country*

^b*Security Research Institute, Placeholder, City, Country*

Abstract

Multi-agent pipelines are increasingly deployed to filter the outputs of large language models (LLMs): a team of LLM agents inspects a candidate response, and a coordinator combines their verdicts into an allow/block decision. Existing designs, exemplified by AutoDefense, sharply reduce jailbreak attack success rates, but they secure the *content* while leaving the *defender itself* unexamined. Such pipelines assume every internal agent is honest, route all verdicts through a single trusted coordinator with no quorum or Byzantine tolerance, feed the untrusted candidate directly to the judges that must evaluate it, are stress-tested only against static attacks, and target jailbreak (harmful content) rather than prompt injection (task-integrity violations that can appear benign). We take the position that a defense pipeline is a multi-agent system whose own security must be analysed, and we study two threats that have no single-model analogue: compromised or colluding *internal* defenders, and *second-order* prompt injection carried by the very content the judges read. We present **Aegis-Agency**, a defense architecture that (i) replaces the single coordinator with Byzantine-robust aggregation of judge verdicts (Krum-style selection, coordinate-wise median, or geometric median), (ii) structurally isolates the untrusted payload from each judge’s instruction channel, (iii) hardens each judge against injection, and (iv) generalises the verdict target from harmful-content classification to instruction-hierarchy / task-integrity. We formalise the semantic verdict space and prove that, under

*Corresponding author.

Email address: `author.one@example.edu` (Author One)

stated honest-concentration and decision-margin assumptions, robust aggregation with $2f+2 < n$ (selection) or $f < n/2$ (median rules) prevents up to f Byzantine judges from flipping the aggregate decision in either direction; we further characterise how the guarantee degrades under correlated judge failures, making the required backbone diversity explicit rather than assumed. We give a security-versus-cost analysis showing token and latency cost linear in the committee size, and an adaptive evaluation protocol covering compromised judges, colluding minorities, second-order payload injection, and attacks on the aggregation rule. We are explicit that a reduced attack success rate under compromise is evidence of graceful degradation, not a proof of end-to-end security; residual failures are reported. The reported empirical figures are placeholders that specify the intended protocol and must be replaced with measured results before submission.

Keywords: LLM security, multi-agent systems, Byzantine-robust aggregation, prompt injection, jailbreak defense, LLM-as-a-judge, insider threat, graceful degradation

1. Introduction

Large language models are now embedded in applications that read untrusted text, call tools, and act on behalf of users. Two failure modes dominate the security discussion. *Jailbreaks* coax a model into producing content its operator intended to withhold [1, 2, 3, 4]. *Prompt injection* is subtler: adversarial instructions embedded in data the model processes—an email, a web page, a tool result—override the operator’s intended task, so that an output that violates the intended instruction hierarchy can nonetheless look benign [5, 6, 7]. Both threats have motivated a wave of defenses, from single-model hardening through preference optimisation and structured inputs [8, 9] to runtime guardrails and moderation classifiers [10, 11, 12, 13].

A distinct and increasingly popular design filters LLM outputs with a *team* of LLM agents rather than a single model. AutoDefense [14] is the canonical example: an Input agent forwards a candidate response to an Intention Analyzer and an Original-Prompt Analyzer, a Judge agent renders a verdict, an optional Llama-Guard agent [10] adds a content check, and a Coordinator combines these into a final allow/block decision enforced by an Output agent. The reported effect is substantial: on one configuration AutoDefense reduces the jailbreak attack success rate from 55.74% to

7.95% using a three-agent LLaMA-2-13b committee [14]. Multi-agent scaffolding of this kind—role specialisation, analysis, and adjudication—mirrors the broader move toward LLM agent societies [15, 16, 17, 18, 19, 20].

The blind spot.. These pipelines secure the content they inspect, but they do not secure *themselves*. The defense is a multi-agent system, and it inherits every threat that multi-agent LLM systems are now known to face [21, 22, 23, 24, 25, 26]. Concretely, an AutoDefense-style agency makes four assumptions that an adversary can attack:

1. **Honest defenders.** Every internal agent is assumed faithful. A single compromised or fine-tuned-to-collude judge, or a supply-chain-tainted analyzer, is outside the model.
2. **A trusted coordinator.** Verdicts flow through one aggregation point with no voting, quorum, or Byzantine tolerance; whoever controls it controls the decision.
3. **Trustworthy self-inspection.** The untrusted candidate is fed directly to the judges. But the judges are themselves LLMs, and text crafted to look like a response can carry instructions aimed at *them*—a *second-order* prompt injection against the defenders, precisely the attack that flips LLM-as-a-judge verdicts [27, 28, 29].
4. **Static adversaries.** Evaluation uses fixed attack sets, not an attacker that adapts to the pipeline or its aggregation rule—exactly the setting where robust aggregators have historically broken [30, 31, 32].

A single hardened model such as SecAlign [8] has no co-defender to collude with and no coordinator to subvert; these threats appear only once the defense is a team. The gap is therefore specific to multi-agent defenses and cannot be closed by swapping in a better single model.

This paper.. We treat the defense pipeline as a first-class attack surface and ask what it takes to tolerate a compromised minority of its own agents while resisting injection from the content it reads. Our answer, **Aegis-Agency**, has four components, illustrated in Fig. 1:

1. *Byzantine-robust verdict aggregation.* The single coordinator is replaced by a robust aggregation rule over judge verdicts—Krum-style selection [33], coordinate-wise median [34], or geometric median [35, 36]—so that a malicious minority cannot flip the decision.

2. *Payload isolation.* The untrusted candidate is confined to a structured data channel, separated from each judge’s instruction channel following the structured-query and spotlighting principles [9, 37], closing the second-order injection surface the defense creates by reading untrusted text.
3. *Hardened judges.* Each judge is individually fine-tuned to resist injection [8, 9], so that isolation and aggregation defend a committee whose members are already robust.
4. *Task-integrity generalisation.* The verdict target is lifted from “harmful content” to “violated the intended task or privilege hierarchy” [7], extending the pipeline from jailbreak filtering to prompt-injection filtering.

What we claim, and what we do not.. We are deliberately conservative. Robust aggregation borrowed from distributed learning comes with a tolerance condition ($2f+2 < n$ for selection [33], $f < n/2$ for median rules [34]), and its guarantees are conditional on the honest verdicts actually concentrating—an assumption that is vacuous if every judge shares a backbone and fails identically [38]. We therefore prove a decision-integrity guarantee *under explicit assumptions*, and separately characterise how it decays as judge failures become correlated. We do not claim that a lower attack success rate under compromise equals security; we claim *graceful degradation* and we report residual failures. Our theoretical results are stated with their assumptions in full, and the empirical numbers in this manuscript are placeholders that fix the evaluation protocol rather than measured outcomes.

Contributions..

- **Threat model (Sections 4 and 5).** The first threat model, to our knowledge within the verified literature, that drops AutoDefense’s honest-agent and single-coordinator assumptions and adds second-order injection against the judges, with attacker knowledge, capability, collusion, and adaptivity made explicit.
- **Architecture (Sections 6 and 7).** Aegis-Agency, combining Byzantine-robust verdict aggregation, payload isolation, and hardened judges, generalised from jailbreak to task-integrity, together with the aggregation algorithm over the semantic verdict space.
- **Theory (Sections 8 and 9).** A decision-integrity theorem for robust verdict aggregation under honest-concentration and margin assumptions,

a correlated-failure analysis that makes the diversity requirement quantitative, and a security-versus-cost complexity analysis.

- **Adaptive evaluation (Section 10).** An evaluation protocol against compromised judges, colluding minorities, second-order payload injection, and adaptive attacks on the aggregation rule—absent from prior multi-agent defenses—with baselines AutoDefense, a single hardened model, and plain majority vote, and a stated correlated-failure test over homogeneous versus diverse backbones.

Section 2 positions the work; Section 3 fixes preliminaries; Sections 4 and 5 give the system and threat model and problem formulation; Sections 6 and 7 present the architecture and algorithm; Sections 8 and 9 develop the theory and cost analysis; Section 10 specifies the evaluation; Sections 11 and 12 discuss implications and limitations; Section 13 concludes.

2. Related Work

We organise prior work into five strands and, for each, state precisely where **Aegis-Agency** departs from it. Figure 2 summarises how the threats we consider map onto the pipeline.

2.1. Multi-agent LLM defenses

AutoDefense [14] is the closest system: a role-specialised committee (Input, Intention Analyzer, Original-Prompt Analyzer, Judge, optional LlamaGuard, Output) coordinated to filter harmful responses, reducing jailbreak attack success rate on a three-agent configuration from 55.74% to 7.95%. Its aggregation is a single Coordinator, its agents are assumed honest, its judges read the untrusted response directly, and its evaluation is static. Runtime guardrail frameworks such as NeMo Guardrails [11] and LlamaFirewall [39] compose several checks around an application, and moderation classifiers [10, 12, 13] provide the individual judges; SmoothLLM [40] aggregates over perturbed copies of a *single* model’s inputs. None of these treats the defender as a set of agents that may include a compromised or colluding minority, and none analyses aggregation under Byzantine faults. **Aegis-Agency** keeps the committee structure but replaces the trusted coordinator with a Byzantine-robust rule and closes the self-inspection surface.

2.2. *Single-model injection and jailbreak hardening*

SecAlign [8] hardens one model against prompt injection by preference optimisation [41]; StruQ [9] separates instructions from data into structured queries and fine-tunes the model to ignore instructions in the data channel; spotlighting [37] marks untrusted spans; the instruction hierarchy [7] trains a model to prioritise privileged instructions. Certified erase-and-check [42] and baseline filters [43] give single-model guarantees against adversarial prompts. These techniques are *components* of **Aegis-Agency** (each judge is hardened, the payload is isolated in StruQ/spotlighting style), and a single hardened model is one of our baselines; they do not, and cannot, address insider compromise of a committee or subversion of an aggregator, which have no single-model analogue.

2.3. *LLM-as-a-judge and its failure modes*

LLM-as-a-judge is now a standard evaluation and adjudication primitive [28], but it inherits biases—position, verbosity, self-enhancement [29]—and can be actively attacked: JudgeDeceiver [27] uses optimisation-based prompt injection to force an LLM judge to select an attacker-chosen answer. This is exactly the second-order injection surface a defense pipeline opens when it feeds untrusted content to its judges. **Aegis-Agency** responds with payload isolation (to remove the lever) and robust aggregation (to bound the effect of the judges an attacker does compromise), and it requires and measures backbone diversity because correlated judge bias is a shared, not independent, failure [38].

2.4. *Security of multi-agent LLM systems*

A growing literature shows that LLM multi-agent systems are attackable through their interactions rather than any single model. Prompt Infection [21] demonstrates self-replicating injections that propagate agent-to-agent; PsySafe [25] and Evil Geniuses [22] show that individual agents can be turned to subvert collective behaviour; Flooding of manipulated knowledge [23] and CORBA [44] show integrity and availability attacks that spread through a community; Agent-in-the-Middle attacks [24] manipulate inter-agent communication; and Agent Security Bench [26] formalises attacks and defenses across agent operation stages. This body of work establishes that insider and communication threats are real for task-performing agent

teams. **Aegis-Agency** is complementary: it studies these threats for a *defensive* team whose explicit job is adjudication, where the natural countermeasure is Byzantine-robust aggregation of verdicts rather than of task outputs.

2.5. Byzantine-robust aggregation and robust statistics

Robust aggregation originates in distributed and federated learning under a *trusted* server [45]. Krum [33] selects the worker gradient closest to its neighbours and tolerates f Byzantine workers when $2f+2 < n$; coordinate-wise median and trimmed mean achieve order-optimal rates under $f < n/2$ [34]; geometric-median aggregation [35, 36] and Bulyan [46] extend the family, the latter warning that single-aggregator defenses can fail in high dimension. The robustness of these estimators rests on classical results—breakdown points of multivariate location estimators [47], geometric-median concentration [48], and high-dimensional robust estimation [49, 50, 51]. Crucially, adaptive attacks break many of these rules: local model poisoning [30], “a little is enough” [31], and inner-product manipulation [32] all evade Krum or median under realistic adversaries, and federated backdoors [52] persist through averaging. We transplant this machinery from gradients-under-a-trusted-master to *verdicts-under-an-untrusted-committee*, which changes both the object (a low-dimensional semantic verdict rather than a high-dimensional gradient) and the adversary (an agent that can also inject the content being judged). The classical Byzantine agreement bounds [53, 54] frame the committee model but are not our mechanism: our aggregation substrate is trusted and non-interactive, so we borrow robust *estimation*, not consensus protocols. Because adaptive attacks are known to defeat these rules, our evaluation is adaptive by construction (Section 10).

Positioning.. To the best of our knowledge, within the verified literature, **Aegis-Agency** is the first defense that (i) models a compromised or colluding minority *inside* an LLM defense pipeline, (ii) replaces its trusted coordinator with Byzantine-robust aggregation over a semantic verdict space, and (iii) closes the second-order injection surface the pipeline creates. We claim graceful degradation under these threats, not provable end-to-end security.

3. Preliminaries and Notation

We collect the notation used throughout; Table 1 is a reference.

Judges and verdicts. A defense agency employs a committee of n *judge* agents indexed by $[n] = \{1, \dots, n\}$. Presented with an inspected payload x (a candidate LLM response and/or an inter-agent message), judge k emits a *verdict*

$$v_k = (d_k, s_k, \mathbf{e}_k), \quad d_k \in \{0, 1\}, \quad s_k \in [0, 1], \quad \mathbf{e}_k \in \mathbb{R}^m, \quad (1)$$

where d_k is the binary decision (0 = allow, 1 = block), s_k is a calibrated block-score (a probability that the payload violates the policy, calibrated in the sense of [55]), and $\mathbf{e}_k$ is a rationale embedding of the judge’s free-text justification produced by a fixed sentence encoder [56, 57]. We stack the continuous part into a *verdict vector*

$$\mathbf{u}_k = (s_k, \mathbf{e}_k) \in \mathbb{R}^d, \quad d = 1 + m, \quad (2)$$

and write $\pi_1(\mathbf{u}_k) = s_k$ for the score coordinate. The decision induced by a verdict vector is $D(\mathbf{u}) = \mathbf{1}\{\pi_1(\mathbf{u}) \geq \tau\}$ for a fixed threshold $\tau \in (0, 1)$.

Honest and Byzantine judges. Let $\mathcal{H} \subseteq [n]$ be the set of *honest* judges and $\mathcal{B} = [n] \setminus \mathcal{H}$ the *Byzantine* (compromised or colluding) judges, with $f = |\mathcal{B}|$ and $|\mathcal{H}| = n - f$. Honest judges execute the intended policy; Byzantine judges may return arbitrary, jointly chosen verdict vectors. We write $\alpha = f/n$ for the Byzantine fraction.

Aggregation. An *aggregation rule* $\mathbf{Agg} : (\mathbb{R}^d)^n \rightarrow \mathbb{R}^d$ maps the n verdict vectors to a single aggregate $\hat{\mathbf{u}} = \mathbf{Agg}(\mathbf{u}_1, \dots, \mathbf{u}_n)$; the agency’s final decision is $\hat{d} = D(\hat{\mathbf{u}})$. We consider three rules from the Byzantine-robust family:

- **Coordinate-wise median** cmed: $\pi_j(\hat{\mathbf{u}}) = \text{median}\{\pi_j(\mathbf{u}_1), \dots, \pi_j(\mathbf{u}_n)\}$ for each coordinate j [34].
- **Geometric median** gmed: $\hat{\mathbf{u}} = \arg \min_{\mathbf{y} \in \mathbb{R}^d} \sum_{k=1}^n \|\mathbf{y} - \mathbf{u}_k\|_2$, computed by a Weiszfeld iteration [35, 36].
- **Krum selection** KRUM: $\hat{\mathbf{u}} = \mathbf{u}_{k^*}$ with $k^* = \arg \min_k \sum_{j \in \mathcal{N}_k} \|\mathbf{u}_k - \mathbf{u}_j\|_2^2$, where $\mathcal{N}_k$ indexes the $n - f - 2$ nearest neighbours of k [33].

Plain **majority vote** aggregates only the decisions, $\hat{d} = \mathbf{1}\{\sum_k d_k > n/2\}$, and is our simplest baseline; it is the sign-vote special case of coordinate-wise median applied to d_k [58].

Table 1: Notation.

Symbol	Meaning
n	number of judge agents in the committee
$f, \alpha = f/n$	number / fraction of Byzantine (compromised or colluding) judges
$\mathcal{H}, \mathcal{B}$	honest and Byzantine judge index sets, $ \mathcal{H} = n - f, \mathcal{B} = f$
x	inspected payload (candidate response and/or inter-agent message)
$v_k = (d_k, s_k, \mathbf{e}_k)$	verdict of judge k : decision, calibrated score, rationale embedding
$\mathbf{u}_k = (s_k, \mathbf{e}_k) \in \mathbb{R}^d$	verdict vector, $d = 1 + m$; $\pi_1(\mathbf{u}_k) = s_k$
τ	block threshold; $D(\mathbf{u}) = \mathbf{1}\{\pi_1(\mathbf{u}) \geq \tau\}$
$\text{Agg}, \hat{\mathbf{u}}$	aggregation rule and aggregate verdict $\hat{\mathbf{u}} = \text{Agg}(\mathbf{u}_1, \dots, \mathbf{u}_n)$
cmed, gmed, KRUM	coordinate-wise median, geometric median, Krum selection
$\mathbf{u}^*$	honest reference verdict (e.g. honest geometric median / consensus)
r	honest concentration radius, $\ \mathbf{u}_k - \mathbf{u}^*\ \leq r$ for $k \in \mathcal{H}$
γ	decision margin, $ \pi_1(\mathbf{u}^*) - \tau \geq \gamma$
C_α	robust-aggregation displacement constant (Lemma 1)
ρ	pairwise correlation of honest-judge errors (Proposition 3)
ε	payload-isolation leakage (Definition 1)

Channels.. Each judge exposes two logically separate inputs: an *instruction channel* carrying the operator’s fixed policy prompt, and a *data channel* carrying the untrusted payload x . *Payload isolation* is the property that instructions appearing in the data channel do not alter the judge’s control flow, realised by structured queries and delimiting [9, 37]; we make this precise in Definition 1.

Norms.. $\|\cdot\|$ denotes the Euclidean norm unless a subscript indicates otherwise. For a finite set S , $|S|$ is its cardinality. $\mathbf{1}\{\cdot\}$ is the indicator.

4. System Model

4.1. The defense agency

Aegis-Agency is a defensive multi-agent system that adjudicates whether a candidate output may be released. It comprises (Fig. 1):

- an **Input agent** that receives the candidate output and any accompanying context and places the untrusted content into a structured data channel;

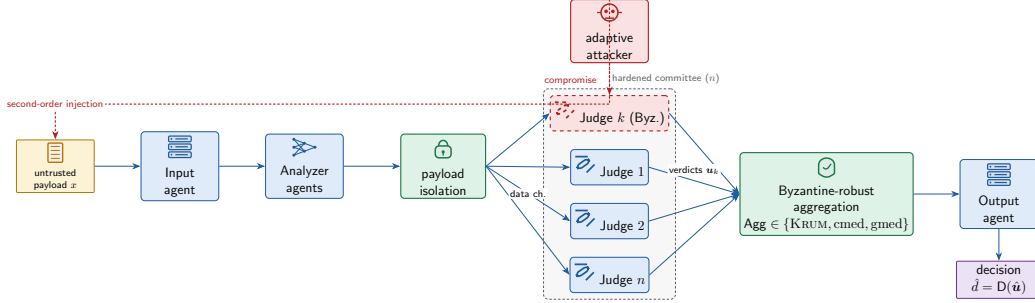


Figure 1: The Aegis-Agency architecture. Blue: honest data/model flow through Input, Analyzer, isolation, the hardened judge committee, robust aggregation, and Output. Green: defensive components (payload isolation and Byzantine-robust verdict aggregation) that replace AutoDefense’s single Coordinator. Red (dashed): the adaptive attacker, which both injects the untrusted payload (second-order injection against the judges) and compromises up to f judges (here Judge k). Judges do not communicate with one another; each sends only its verdict u_k to the aggregator. Icons are pure TikZ. Arrow lanes are separated by colour and dash pattern; labels sit on white chips so no connector crosses text.

- one or more **Analyzer agents** that extract task-relevant features—the apparent user intent and the original instruction/privilege hierarchy [7]—and attach them to the instruction channel as trusted metadata;
- a committee of n **Judge agents** ($1 \leq n \leq 7$ in our study), each a hardened LLM [8, 9] that reads the isolated payload and emits a verdict v_k as in Eq. (1);
- a **robust aggregator** that replaces AutoDefense’s single Coordinator with a Byzantine-robust rule **Agg** over $\{u_k\}$; and
- an **Output agent** that enforces the aggregate decision $\hat{d}$ (release, block, or escalate).

The committee is deployed in one of two topologies. In the *parallel* topology the n judges are queried independently and concurrently; in the *sequential* topology they are queried in series (relevant only to latency, Section 9). Judges do not exchange messages with one another; each communicates only with the aggregator. This star topology is deliberate: it removes the inter-agent communication channel that recent attacks exploit [24, 21] and confines adjudication to verdicts the aggregator can robustly combine.

4.2. Trust boundaries

Three assets are *trusted*: the aggregation substrate executes **Agg** correctly and non-interactively; the instruction channel delivering the operator’s policy is authentic; and at least $n - f$ judges are honest, with f small enough to satisfy the tolerance condition of the chosen rule ($2f+2 < n$ for Krum, $f < n/2$ for median rules). Two assets are *untrusted*: the payload x , which may be adversarially crafted, and up to f judges, which may be compromised or colluding. The aggregator being trusted is what distinguishes our setting from Byzantine agreement among mutually distrustful parties [53, 54]: we need robust *estimation* of a verdict, not a consensus protocol. This mirrors the trusted-server assumption of robust federated aggregation [33, 34], transplanted from gradients to verdicts.

4.3. Threat model

We adopt the standard security-analysis structure of goal, knowledge, and capability.

Attacker goal. Two objectives, dual to each other:

- *Integrity violation*: cause the agency to **pass** an output that is unsafe or that violates the intended task/instruction hierarchy—an evasion, measured by attack success rate under compromise.
- *Availability violation*: cause the agency to **block** benign outputs—over-refusal, a denial-of-service against legitimate traffic [44].

Attacker knowledge. Black-box to gray-box. The attacker may know the pipeline design, the aggregation rule, the judges’ backbones, and the policy prompt, but not the operator’s private randomness (e.g. any per-request nonce used for isolation delimiting). This spans the practical range from a third party who only submits payloads to an insider who fine-tuned a judge.

Attacker capabilities. Three levers, usable in combination:

1. **Insider compromise.** Control up to f judges, which then return arbitrary, jointly optimised verdict vectors. This is the LLM-committee analogue of Byzantine workers [33] and of poisoned federated clients [52, 30].
2. **Second-order injection.** Craft the payload x so that, in addition to being the object under review, it carries instructions aimed at the honest judges reading it [27, 5]. Without isolation this can flip honest verdicts; with isolation it is reduced to the leakage ε of Definition 1.

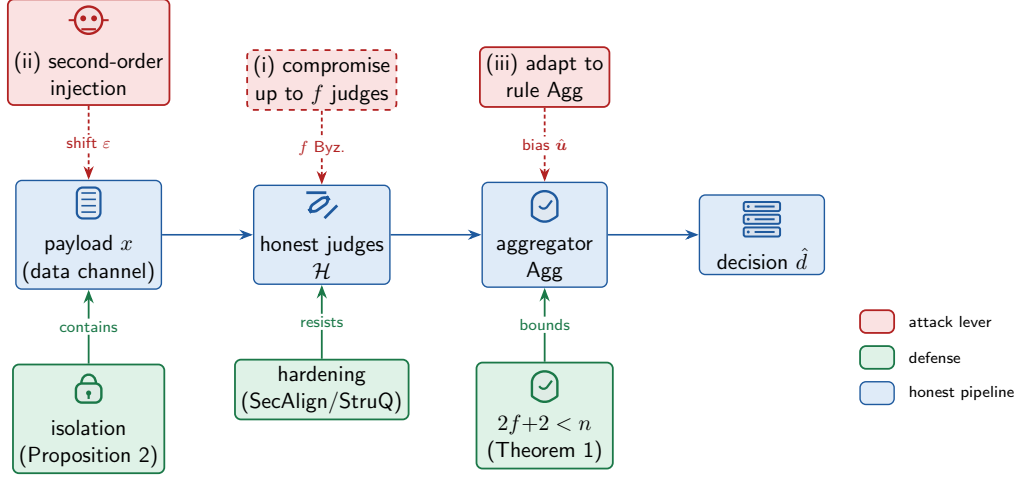


Figure 2: Threat-and-defense map. The honest pipeline spine (blue) runs left to right. Above it (red), the three composable attacker levers: (i) compromising up to f judges (dashed Byzantine node), (ii) second-order injection of the payload, and (iii) adapting to the aggregation rule. Below it (green), the mechanism that contains each lever: isolation for (ii), judge hardening for the judges, and the $2f+2 < n$ robust-aggregation guarantee for (iii). Each attack and defense connects to the single component it acts on; lanes are separated by colour and dash, and every label sits on a white chip.

3. **Adaptivity.** Optimise the attack against the specific aggregation rule, as adaptive attacks do against robust federated aggregators [30, 31, 32]. Colluding judges may coordinate to stay within the honest verdict spread while biasing the aggregate.

Collusion.. Byzantine judges are not independent: they may share a strategy and coordinate their verdict vectors to move the aggregate as far as the tolerance condition permits. Our analysis and evaluation treat collusion as the default, not the worst case of an otherwise-independent fault.

Out of scope.. We do not defend against (i) all judges compromised ($f \geq n - f$), which no minority-tolerance rule can address; (ii) a correlated failure that affects *every* judge identically—studied in Section 8 as a limiting case that voids the guarantee, and mitigated but not eliminated by backbone diversity; (iii) compromise of the trusted aggregation substrate or the instruction channel; and (iv) attacks on the underlying serving infrastructure. Figure 2 maps each in-scope lever to the component it targets and the defense that contains it.

5. Problem Formulation

5.1. The adjudication task

Let x be an inspected payload drawn from a distribution over benign and adversarial content, and let $y^*(x) \in \{0, 1\}$ be the ground-truth label (1 if x must be blocked because it is unsafe or violates the intended task/instruction hierarchy, 0 otherwise). The agency computes verdicts $\{\mathbf{u}_k\}_{k \in [n]}$, aggregates them to $\hat{\mathbf{u}} = \text{Agg}(\mathbf{u}_1, \dots, \mathbf{u}_n)$, and outputs $\hat{d} = D(\hat{\mathbf{u}})$. We evaluate the agency by two dual error rates, each measured under an adversary that controls up to f judges and the payload:

$$\text{ASR-UC}(f) = \mathbb{P}[\hat{d} = 0 \mid y^*(x) = 1] \quad (\text{attack success under compromise / evasion}), \quad (3)$$

$$\text{ORR}(f) = \mathbb{P}[\hat{d} = 1 \mid y^*(x) = 0] \quad (\text{over-refusal / availability loss}). \quad (4)$$

The probabilities are over the payload distribution, the judges' stochasticity, and the adversary's choices. ASR-UC generalises the jailbreak attack success rate of AutoDefense [14] to (i) task-integrity violations and (ii) the presence of compromised judges; ORR captures the cost the idea flags explicitly, that a block-biased aggregator harms benign traffic.

5.2. Design goals

(G1) Decision integrity under a Byzantine minority. For f below the rule's tolerance, up to f colluding judges must not be able to force $\hat{d} = 0$ on a payload the honest committee would block, nor $\hat{d} = 1$ on one it would allow. This is the property flagged in the idea as "to be proven, not assumed"; Section 8 establishes it under explicit assumptions.

(G2) Second-order injection containment. Instructions embedded in the payload must not, on their own, change honest verdicts; the attacker's only remaining lever should be to spend from its budget of f compromised judges.

(G3) Graceful degradation, not brittleness. As f grows toward the tolerance boundary and as honest verdicts become correlated, ASR-UC and ORR should rise smoothly and predictably, with the dependence characterised rather than assumed away.

(G4) Bounded cost. The security gain must be accounted against latency and token/API cost, both of which grow with the committee size n .

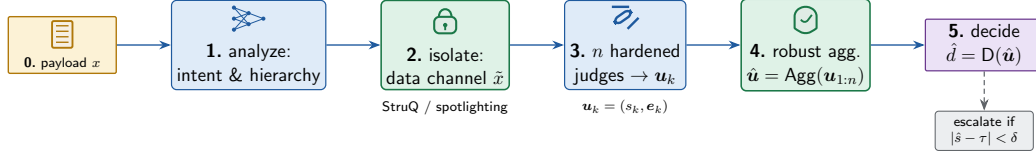


Figure 3: The **Aegis-Agency** method pipeline as five numbered, left-to-right stages (Algorithm 1). Stage 2 (isolation, green) and stage 4 (robust aggregation, green) are the defensive additions; stage 3 (blue) is the hardened judge committee. The control-flow branch (gray, dashed) escalates low-margin cases to review. Short equations are placed on white chips beneath their stage so no arrow crosses text.

5.3. What “secure” does and does not mean here

We state a non-claim precisely, because it governs the theory and the evaluation. A reduction in $\text{ASR-UC}(f)$ relative to a single model or a plain majority vote is *evidence of graceful degradation*. It is *not* a proof that the agency is secure: our guarantees are conditional on assumptions (honest concentration, decision margin, tolerable correlation) that may fail in deployment, and adaptive attacks are known to defeat robust aggregators when their assumptions are violated [30, 31]. We therefore (a) prove (G1)–(G2) under assumptions we state in full, (b) characterise the degradation in (G3) quantitatively, and (c) report residual failures and the security-versus-cost frontier (G4) rather than a single headline number.

5.4. Threat-to-defense map

Table 2 pins each attacker lever to the design goal and mechanism that addresses it, and to where it is analysed and tested. The rest of the paper follows this map.

6. The Aegis-Agency Architecture

Aegis-Agency instantiates the four components of Section 1 as a single pipeline (Fig. 3). We describe each, then the design rationale that ties them together.

6.1. Payload isolation

The Input agent places the untrusted candidate into a structured data channel that is lexically and positionally separated from the judges’ instruction channel, following the structured-query design of StruQ [9] and the marking discipline of spotlighting [37]. Concretely, the payload is wrapped in reserved, per-request delimiters unknown to the attacker and never interpreted

Table 2: Threat-to-defense map. Each attacker lever is contained by a specific mechanism, analysed under specific assumptions, and exercised by a specific attack in the evaluation.

Attacker lever	Design goal	Mechanism	Analysed / tested by
Compromise up to f judges	(G1) integrity	Byzantine-robust aggregation (KRUM/cmed/gmed)	Theorem 1; compromised-judge & colluding attacks
Second-order injection of payload	(G2) containment	payload isolation (structured data channel) + hardened judges	Definition 1, Proposition 2; second-order injection attack
Adapt to the aggregation rule	(G1),(G3)	tolerance condition + diversity; no reliance on a single aggregator	Theorem 1, Proposition 3; adaptive-on-aggregation attack
Force over-refusal (availability)	(G1) availability	symmetric robust-aggregation guarantee	Corollary 1; over-refusal measurement
Correlated judge failure	(G3) degradation	backbone/prompt diversity; measured residual correlation	Proposition 3; homogeneous-vs-diverse ablation

as control tokens, and the judge is fine-tuned to treat everything in the data channel as inert evidence to be classified, not as instructions to follow. We formalise the target property:

Definition 1 (ε -isolation). A judge with verdict distribution $P(\cdot \mid c, x)$ given instruction-channel content c and data-channel payload x is ε -isolated if for every payload x and every “injected” variant x' that differs from x only by added instruction-like content,

$$\sup_c \text{TV}(P(\cdot \mid c, x), P(\cdot \mid c, x')) \leq \varepsilon, \quad (5)$$

where TV is total-variation distance over the verdict v . Perfect isolation is $\varepsilon = 0$.

ε is an empirical property of a hardened, isolated judge, measured in Section 10 by the shift in verdicts induced by JudgeDeceiver-style injections [27].

Isolation does not make a judge correct; it makes its verdict insensitive to instructions hidden in what it reads, which is what (G2) requires.

6.2. Hardened judges

Each judge is an LLM fine-tuned for injection resistance by preference optimisation [8, 41] and structured-input training [9], on top of an aligned backbone [59, 60, 61]. A judge outputs the verdict of Eq. (1): a decision, a score calibrated by temperature scaling [55] so that scores are comparable across judges, and a rationale embedding from a fixed sentence encoder [56, 57] that places the free-text justification in a shared semantic space. The embedding is what lets robust aggregation see *how* a judge reasoned, not only *what* it decided, so that a colluding judge that reports the honest decision but a manipulated rationale is still an outlier in $\mathbb{R}^d$.

6.3. Byzantine-robust verdict aggregation

The aggregator replaces the single Coordinator with a robust rule **Agg** over $\{\mathbf{u}_k\}$. We support the three rules of Section 3. cmed and gmed tolerate $f < n/2$ and are cheap; KRUM tolerates f with $2f+2 < n$ and returns an *actual* judge verdict (useful when the operator wants an attributable rationale). The aggregate verdict $\hat{\mathbf{u}}$ yields the decision $\hat{d} = D(\hat{\mathbf{u}})$. The design choice that makes this meaningful is aggregating in the *semantic verdict space* $\mathbb{R}^d$ rather than over decisions alone: a plain majority vote sees only d_k and is blind to a colluder that votes with the honest majority while skewing scores to shift a downstream threshold or to poison an audit log; aggregating $(s_k, \mathbf{e}_k)$ exposes such a colluder as a geometric outlier.

6.4. Task-integrity generalisation

The verdict target is lifted from “is this content harmful?” to “does this output violate the intended task or privilege hierarchy?” [7]. The Analyzer agents supply the honest task specification on the instruction channel; each judge decides whether the payload, treated as inert data, is consistent with that specification. This is what turns a jailbreak filter into a prompt-injection filter: an output can be perfectly harmless in isolation yet still constitute a successful injection if it obeys an instruction that was smuggled through the data channel rather than issued by the operator [5, 6].

6.5. Design rationale: why all four are needed

The components are not independent add-ons; each closes a gap the others leave open, and this is the core of our novelty argument.

- *Hardened judges alone* are the single-model baseline (SecAlign [8]). They have no co-defender to collude and no aggregator to subvert—so they cannot express, let alone defend, the insider threat.
- *Robust aggregation alone*, over un-isolated judges, is defeated by second-order injection: an attacker who can flip honest verdicts through the data channel moves the honest majority itself, and no minority-tolerance rule helps once the honest inputs are corrupted [27].
- *Isolation alone*, over a single un-aggregated judge, still falls to a single compromised judge, because there is no committee.
- *Aggregation over decisions alone* (plain majority vote) is blind to score/rationale manipulation and, as Proposition 3 shows, collapses under correlated failure.

The combination gives (G1) from aggregation, (G2) from isolation plus hardening, and (G3) from the diversity requirement that aggregation’s assumptions impose. Figure 3 shows the flow; Section 7 gives the procedure; Section 8 proves what the combination buys and states precisely where it stops.

7. Algorithm

Algorithm 1 states the Aegis-Agency adjudication procedure, and Algorithm 2 the robust aggregation subroutines. The pipeline is a single non-interactive pass: analyze, isolate, query n hardened judges (in parallel), aggregate their verdict vectors robustly, and enforce the decision. No judge sees another judge’s verdict, so a compromised judge cannot influence honest ones; its only reach is the aggregator, which is robust by construction.

Remarks on the algorithm.. The *escalate* branch (Algorithm 1, line 11) implements graceful degradation at the decision level: when the robust aggregate lands within δ of the threshold—the regime where Theorem 1 cannot certify the decision—the agency defers rather than guesses, converting an uncertain automated verdict into a review request. Setting $\delta = 0$ recovers a pure allow/block agency. The three subroutines expose the security–cost trade-off of Section 9: *cmed* is $O(nd)$ per request (with an $O(n \log n)$ selection per coordinate), *gmed* is $O(Tnd)$ for T Weiszfeld iterations, and *KRUM* is $O(n^2d)$

Algorithm 1 Aegis-Agency adjudication

Require: payload x ; policy prompt c_0 ; committee of n hardened judges $\{J_k\}$; aggregation rule $\text{Agg} \in \{\text{KRUM}, \text{cmed}, \text{gmed}\}$; threshold τ ; encoders (score calibrator g , sentence encoder ϕ)

Ensure: decision $\hat{d} \in \{\text{allow}, \text{block}, \text{escalate}\}$

- 1: $(\text{intent}, \text{hierarchy}) \leftarrow \text{ANALYZE}(x)$ $\triangleright$ trusted task spec on instruction channel
 - 2: $c \leftarrow c_0 \parallel \text{intent} \parallel \text{hierarchy}$
 - 3: $\tilde{x} \leftarrow \text{ISOLATE}(x)$ $\triangleright$ wrap payload in per-request delimiters; data channel only
 - 4: **for** $k = 1$ **to** n **in parallel do**
 - 5: $(d_k, \text{raw}_k, \text{rationale}_k) \leftarrow J_k(\text{instruction}=c, \text{data}=\tilde{x})$
 - 6: $s_k \leftarrow g(\text{raw}_k)$; $\mathbf{e}_k \leftarrow \phi(\text{rationale}_k)$ $\triangleright$ calibrated score; rationale embedding
 - 7: $\mathbf{u}_k \leftarrow (s_k, \mathbf{e}_k) \in \mathbb{R}^d$
 - 8: **end for**
 - 9: $\hat{\mathbf{u}} \leftarrow \text{Agg}(\mathbf{u}_1, \dots, \mathbf{u}_n)$ $\triangleright$ robust aggregation in verdict space (Algorithm 2)
 - 10: $\hat{s} \leftarrow \pi_1(\hat{\mathbf{u}})$
 - 11: **if** $|\hat{s} - \tau| < \delta$ **then return escalate** $\triangleright$ low-margin case: defer to human/secondary review
 - 12: **else if** $\hat{s} \geq \tau$ **then return block**
 - 13: **else return allow**
 - 14: **end if**
-

from its pairwise distances. All three are negligible beside the n LLM calls, which dominate cost and are the reason n is kept small.

8. Theoretical Analysis

We establish the decision-integrity property (G1) for robust verdict aggregation under explicit assumptions, its availability dual, the second-order-injection containment (G2), and the correlated-failure degradation (G3). Proofs of the two results marked with a dagger ($\dagger$) are complete and given in Appendix Appendix A; the remaining statements are labelled as proof sketches or as restatements of cited results, and none is presented as more than it is. Figure 4 shows the dependency structure.

Algorithm 2 Robust aggregation subroutines $\text{Agg}(\mathbf{u}_1, \dots, \mathbf{u}_n)$

```

1: function COORDINATEMEDIAN( $\mathbf{u}_1, \dots, \mathbf{u}_n$ )
2:   for  $j = 1$  to  $d$  do
3:      $\pi_j(\hat{\mathbf{u}}) \leftarrow \text{median}\{\pi_j(\mathbf{u}_1), \dots, \pi_j(\mathbf{u}_n)\}$ 
4:   end for
5:   return  $\hat{\mathbf{u}}$ 
6: end function

7: function GEOMETRICMEDIAN( $\mathbf{u}_1, \dots, \mathbf{u}_n$ ; tolerance  $\eta$ )
8:    $\mathbf{y} \leftarrow \frac{1}{n} \sum_k \mathbf{u}_k$  ▷ Weiszfeld iteration [36]
9:   repeat
10:     $w_k \leftarrow 1 / \max(\|\mathbf{y} - \mathbf{u}_k\|, \eta)$  for each  $k$ 
11:     $\mathbf{y} \leftarrow (\sum_k w_k \mathbf{u}_k) / (\sum_k w_k)$ 
12:   until convergence
13:   return  $\mathbf{y}$ 
14: end function

15: function KRUM( $\mathbf{u}_1, \dots, \mathbf{u}_n$ ; assumed faults  $f$ )
16:   for  $k = 1$  to  $n$  do
17:      $\mathcal{N}_k \leftarrow$  indices of the  $n - f - 2$  nearest  $\mathbf{u}_j$  ( $j \neq k$ ) to  $\mathbf{u}_k$ 
18:      $\text{score}(k) \leftarrow \sum_{j \in \mathcal{N}_k} \|\mathbf{u}_k - \mathbf{u}_j\|^2$ 
19:   end for
20:   return  $\mathbf{u}_{k^*}$  where  $k^* = \arg \min_k \text{score}(k)$ 
21: end function

```

8.1. Assumptions

Assumption 1 (Honest concentration). There is a reference verdict $\mathbf{u}^* \in \mathbb{R}^d$ and a radius $r \geq 0$ such that every honest judge's verdict vector satisfies $\|\mathbf{u}_k - \mathbf{u}^*\| \leq r$ for all $k \in \mathcal{H}$.

Assumption 2 (Decision margin). The reference score is bounded away from the threshold: $|\pi_1(\mathbf{u}^*) - \tau| \geq \gamma$ for some margin $\gamma > 0$, and the reference decision equals the ground truth, $D(\mathbf{u}^*) = y^*(x)$.

Assumption 3 (Tolerable Byzantine fraction). For median-type rules (cmed, gmed), $f < n/2$; for KRUM, $2f+2 < n$.

Remark 1. Assumptions 1 and 2 are the substantive assumptions and they are *empirical claims about honest judges*, not mathematical conveniences.

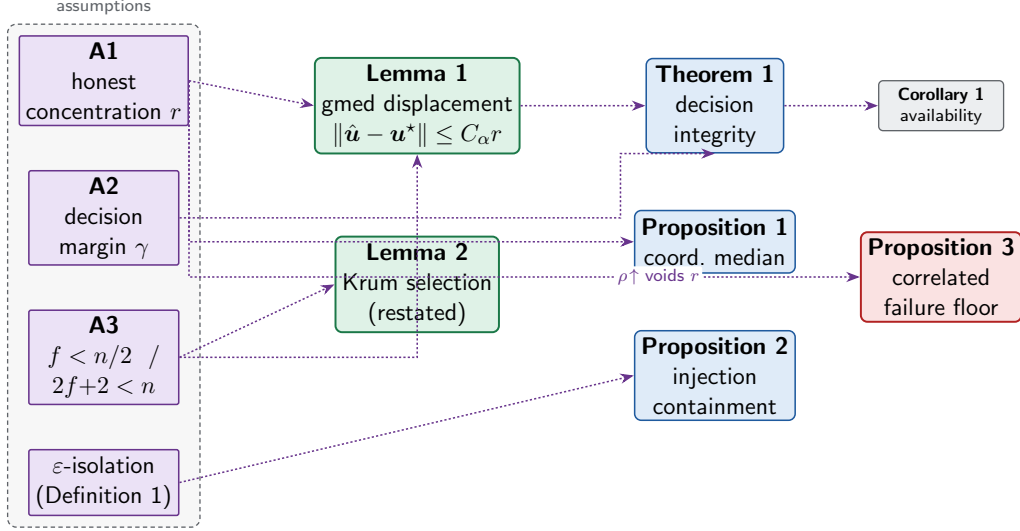


Figure 4: Theory dependency graph. Assumptions (purple) feed lemmas (green), which feed theorems and propositions (blue); the availability corollary (gray) and the correlated-failure floor (red, a limiting-case impossibility that *voids* the concentration assumption) are downstream. Routed orthogonal connectors avoid diagonal clutter; the only annotated edge notes that rising error correlation ρ invalidates the honest radius r that Lemma 1 and Theorem 1 require.

Honest concentration can fail if honest judges genuinely disagree (a hard, ambiguous payload) or if they share a systematic bias [28, 29]; the margin can be small near policy boundaries. We measure r and γ in Section 10 rather than assume favourable values, and Proposition 3 treats the case where concentration degrades because honest errors are correlated.

8.2. Robustness of the aggregate verdict

The engine of (G1) is a deterministic robustness bound for the geometric median. We state it and give the complete proof in Appendix Appendix A.

Lemma 1 (Geometric-median displacement[†]). *Let $\mathbf{u}_1, \dots, \mathbf{u}_n \in \mathbb{R}^d$, and suppose a subset $\mathcal{H}$ with $|\mathcal{H}| = n - f$ satisfies $\|\mathbf{u}_k - \mathbf{u}^*\| \leq r$ for all $k \in \mathcal{H}$. If $f < n/2$, then the geometric median $\hat{\mathbf{u}} = \text{gmed}(\mathbf{u}_1, \dots, \mathbf{u}_n)$ satisfies*

$$\|\hat{\mathbf{u}} - \mathbf{u}^*\| \leq C_\alpha r, \quad C_\alpha = \frac{2(n-f)}{n-2f} = \frac{2(1-\alpha)}{1-2\alpha}, \quad (6)$$

where $\alpha = f/n < \frac{1}{2}$. The bound holds for arbitrary (adversarially chosen) Byzantine verdicts $\{\mathbf{u}_k\}_{k \in \mathcal{B}}$.

Remark 2. C_α is finite and increasing on $[0, \frac{1}{2})$, diverging as $\alpha \rightarrow \frac{1}{2}$: $C_0 = 2$, $C_{1/4} = 3$, $C_{1/3} = 4$. Thus tolerance is graceful, not binary—the aggregate can be pushed further as the Byzantine fraction approaches one half, exactly the graceful degradation (G3) asks for. The constant matches the breakdown behaviour of multivariate location estimators [47, 48]. We are careful *not* to claim this constant is optimal or that it is a matching bound; it is a sufficient displacement bound.

8.3. Decision integrity and availability

Theorem 1 (Decision integrity under a Byzantine minority[†]). *Under Assumptions 1 and 2 and Assumption 3 with a median-type rule, if the margin dominates the displacement,*

$$\gamma > C_\alpha r, \quad (7)$$

then the aggregate decision equals the honest reference decision, $D(\hat{\mathbf{u}}) = D(\mathbf{u}^) = y^*(x)$, for every choice of the f Byzantine verdicts. Consequently up to f colluding judges can neither force a block payload to pass nor a benign payload to be blocked.*

Corollary 1 (Availability). *Under the hypotheses of Theorem 1 the over-refusal event $\{\hat{d} = 1 \mid y^* = 0\}$ cannot be induced by the f Byzantine judges: the guarantee is symmetric in the two decision directions. Hence robust aggregation addresses the availability goal of Section 5 on the same terms as the integrity goal.*

Theorem 1 is the precise form of the property the idea flagged as “to be proven, not assumed.” Its content is conditional: robust aggregation converts a minority of arbitrary Byzantine verdicts into a bounded displacement of the aggregate, and a sufficient decision margin converts a bounded displacement into an unchanged decision. The condition $\gamma > C_\alpha r$ is where the empirical burden sits—it is satisfied only if honest judges concentrate tightly (r small) relative to their distance from the policy boundary (γ large), which is precisely what backbone diversity and hardening are meant to deliver and what Section 10 measures.

8.4. Coordinate-wise median and Krum

Proposition 1 (Coordinate-median integrity). *Under Assumptions 1 and 3 with $f < n/2$, each coordinate of $\hat{\mathbf{u}} = \text{cmcd}(\mathbf{u}_1, \dots, \mathbf{u}_n)$ lies within the range*

of the honest coordinates: $\min_{k \in \mathcal{H}} \pi_j(\mathbf{u}_k) \leq \pi_j(\hat{\mathbf{u}}) \leq \max_{k \in \mathcal{H}} \pi_j(\mathbf{u}_k)$ for all j . In particular $|\pi_1(\hat{\mathbf{u}}) - \pi_1(\mathbf{u}^*)| \leq r$, and Eq. (7) with C_α replaced by 1 suffices for Theorem 1 to hold for the score coordinate.

Proof sketch. With $f < n/2$ Byzantine values among n , the median order statistic of each coordinate is bracketed by two honest order statistics, hence lies in the honest range; the score-coordinate bound and the margin argument then follow as in Theorem 1. The full argument is a one-dimensional specialisation and is given in Appendix A. This recovers the tolerance behaviour of coordinate-wise median in Byzantine learning [34]. $\square$

Lemma 2 (Krum selection, restated from [33]). *Under Assumption 3 with $2f+2 < n$, the vector selected by KRUM has, in expectation over honest stochasticity, squared distance to the honest barycenter bounded by a constant multiple of the honest variance; in particular a Byzantine minority cannot force KRUM to select a vector arbitrarily far from the honest verdicts.*

Remark 3. Lemma 2 is a restatement of the guarantee of Blanchard et al. [33] adapted to our verdict space; we do not reprove it. Two caveats are transplanted with it. First, the guarantee bounds distance to the honest barycenter, so it feeds Theorem 1 only under Assumption 1 with a possibly larger effective r . Second, single-selection rules such as KRUM can behave poorly in high dimension [46]; because our verdict space is low-dimensional after embedding compression ($d = 1 + m$ with modest m), and because cmed/gmed avoid the single-selection pathology, we treat KRUM as one option among three rather than the default.

8.5. Second-order injection containment

Proposition 2 (Isolation containment). *Suppose every honest judge is ε -isolated (Definition 1) and let a second-order injection alter the payload from x to x' . Then the injection changes the decision of any fixed honest judge with probability at most ε , and by a union bound it changes more than t honest decisions with probability at most $\binom{n-f}{t+1} \varepsilon^{t+1}$ under independent judge draws. In the limit $\varepsilon \rightarrow 0$ (perfect isolation), a second-order injection that does not compromise a judge's parameters leaves all honest verdicts unchanged, so the attacker's only remaining lever is to spend from its budget of f compromised judges, which is governed by Theorem 1.*

Proof sketch. The per-judge bound is immediate from Definition 1: total-variation distance ε bounds the change in probability of any event, including

the decision event. The union/count bound assumes honest judges respond independently to the injection, which holds when their randomness is independent; correlated responses are exactly the regime of Proposition 3. The $\varepsilon \rightarrow 0$ statement is the definition of perfect isolation. See Appendix Appendix A. $\square$

Remark 4. Proposition 2 is conditional on the measured leakage ε ; it does not assert $\varepsilon = 0$ in practice. Its role is to separate two attack channels that un-isolated pipelines conflate: without isolation, a payload injection can move honest verdicts *for free* (outside the Byzantine budget), defeating any minority-tolerance rule; with isolation, injection is priced into the budget f that Theorem 1 already handles. This is the formal content of the claim that isolation “closes the surface the defense creates by reading untrusted text.”

8.6. Correlated failure: when the guarantee is vacuous

Assumption 1 presumes honest judges cluster near $\mathbf{u}^*$. If they instead fail *together*, the honest inputs to the aggregator are themselves wrong and no robust rule can recover the correct decision. We make the dependence on correlation quantitative for the decision-level (majority) view, which upper-bounds what any rule using only decisions can achieve.

Proposition 3 (Correlated honest failure). *Model the honest judges’ correctness indicators $\{Z_k\}_{k \in \mathcal{H}}$, $Z_k = \mathbf{1}\{J_k \text{ decides correctly}\}$, as exchangeable Bernoulli variables with mean $\mu = \mathbb{P}[Z_k = 1]$ and pairwise correlation $\rho \in [0, 1]$. Let $\bar{Z} = \frac{1}{n-f} \sum_{k \in \mathcal{H}} Z_k$ be the honest correct fraction. Then*

$$\mathbb{E}[\bar{Z}] = \mu, \quad \text{Var}(\bar{Z}) = \mu(1 - \mu) \left[\frac{1 - \rho}{n - f} + \rho \right], \quad (8)$$

so as $n \rightarrow \infty$, $\text{Var}(\bar{Z}) \rightarrow \mu(1 - \mu)\rho > 0$ for any $\rho > 0$. Consequently, if $\mu < \frac{1}{2}$ (honest judges are more often wrong than right on a class of payloads), adding judges does not drive the probability of a correct honest majority to one; and for $\rho > 0$ the honest majority fails with probability bounded below by a constant independent of n (Chebyshev, Appendix Appendix A). Backbone/prompt diversity is the mechanism that reduces ρ ; the ceiling in Eq. (8) is why diversity is a requirement, not an optimisation.

Table 3: Guarantees, assumptions, and proof status. “Complete” proofs appear in Appendix Appendix A; “restated” results are cited, not reproven; “sketch” proofs give the argument with the routine specialisation deferred to the appendix.

Result	Assumptions	Goal	Status
Lemma 1 (gmed displacement)	concentration; $f < n/2$	(G1)	complete
Theorem 1 (integrity)	concentration; margin $\gamma > C_\alpha r$; $f < n/2$	(G1)	complete
Corollary 1 (availability)	as Theorem 1	(G1) avail.	complete (corollary)
Proposition 1 (co-ordinate median)	concentration; $f < n/2$	(G1)	sketch
Lemma 2 (Krum)	$2f+2 < n$	(G1)	restated [33]
Proposition 2 (isolation)	ε -isolation	(G2)	sketch
Proposition 3 (correlated failure)	exchangeable honest errors, correlation ρ	(G3)	complete (limiting case)

Remark 5. Proposition 3 formalises the idea’s warning that “Krum tolerance is vacuous if all judges fail identically” ($\rho \rightarrow 1$ collapses the committee to a single effective judge) and that diversity is needed to make the committee more than a single point of failure. It also explains why plain majority vote and any verdict-space rule share the same failure mode at the decision level: robustness to a Byzantine *minority* cannot substitute for correctness of the honest *majority*. This is a limiting-case analysis, not a defense; it is exactly the residual failure the idea insists on reporting.

8.7. Summary of guarantees

Table 3 collects the results, their assumptions, and their status. The two complete proofs (Lemma 1 and Theorem 1) are what establish (G1); the remaining statements support (G2), the median/Krum variants, and the honest boundary of the whole approach.

9. Complexity and the Security–Cost Frontier

Aegis-Agency buys robustness by running a committee, and the idea is explicit that the cost of that committee may be unattractive at small n . We

make the cost quantitative and pair it with the tolerance it purchases, so the trade-off is stated rather than hidden.

9.1. Token and API cost

Each request issues one hardened-judge call per committee member plus the Analyzer calls. Let ℓ_{in} and ℓ_{out} be the per-judge input and output token counts. The token cost per adjudication is

$$\text{Tokens}(n) = n(\ell_{\text{in}} + \ell_{\text{out}}) + \text{Tokens}_{\text{analyze}}, \quad (9)$$

i.e. *linear* in the committee size n , since judges are independent and the payload is broadcast unchanged. Payload isolation adds only the fixed delimiter overhead. This linearity is the cost the idea asks to quantify, and it is the reason n is restricted to the range 1–7: the token bill grows in lockstep with n while, by Lemma 1, the tolerated fault count grows only as $\lfloor (n-1)/2 \rfloor$ (median rules) or $\lfloor (n-2)/2 \rfloor$ (Krum).

9.2. Latency

In the parallel topology the judge calls overlap, so wall-clock latency is

$$\text{Lat}_{\parallel}(n) = \max_{k \in [n]} \text{lat}(J_k) + \text{lat}(\mathbf{Agg}) + \text{lat}_{\text{analyze}}, \quad (10)$$

dominated by the slowest judge plus a small aggregation term; in the sequential topology it is $\sum_k \text{lat}(J_k) + \text{lat}(\mathbf{Agg})$, linear in n . The idea notes latency is “linear in the number of agents”; Eq. (10) shows this is the *sequential* worst case and that parallel deployment reduces the judge contribution to a constant in n (at unchanged token cost), trading hardware concurrency for latency.

9.3. Aggregation cost

The three rules cost, per request, $O(nd)$ with an $O(n \log n)$ per-coordinate selection for cmed; $O(Tnd)$ for gmed with T Weiszfeld iterations [36]; and $O(n^2d)$ for KRUM from its pairwise distances [33]. With $n \leq 7$ and modest embedding dimension $d = 1 + m$, all three are dominated by the LLM calls of Eq. (9) and contribute negligibly to latency.

Table 4: Security–cost frontier for median-type aggregation. Tolerance is the maximum f with $f < n/2$; token cost is the multiple of a single-judge call from Eq. (9) (Analyzer overhead excluded). Small committees tolerate very few faults, the feasibility risk the idea flags as an open empirical question.

Committee n	Max tolerated f	Tolerance fraction α^*	Token cost ($\times$ single judge)
1	0	0.00	1
2	0	0.00	2
3	1	0.33	3
4	1	0.25	4
5	2	0.40	5
6	2	0.33	6
7	3	0.43	7

9.4. The frontier

Define the *Byzantine tolerance fraction* $\alpha^*(n) = \lfloor (n-1)/2 \rfloor / n$ for median rules. Table 4 tabulates tolerance against the multiplicative token cost n . The security gain is discrete and, at small n , coarse: moving from $n = 3$ (tolerates $f = 1$) to $n = 5$ (tolerates $f = 2$) doubles neither cost-effectiveness nor tolerance cleanly. Whether a given operating point is worthwhile is an empirical question about the deployment’s threat level and latency budget, which Section 10 is designed to answer rather than to presuppose.

10. Experimental Design

This section specifies the evaluation protocol. **The numerical entries in Tables 5 and 6 and the curves in Fig. 6 are placeholders that encode expected trends and fix the measurement procedure; they are not experimental results and must be replaced with measured values before submission.** No superiority claim is made on their basis. Figure 5 summarises the design.

10.1. Research questions

RQ1 (integrity under compromise). As the compromised/colluding fraction f/n grows to the tolerance boundary, how does ASR-UC of **Aegis-Agency** compare with AutoDefense (single coordinator), a single hardened model, and plain majority vote?

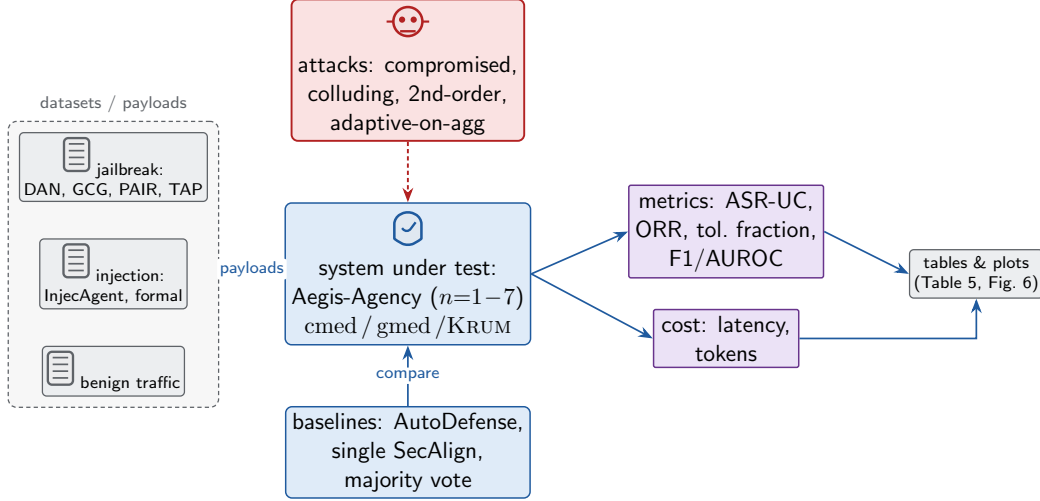


Figure 5: Experimental setup (Section 10). Datasets and benign traffic (gray) feed the system under test, **Aegis-Agency** with committee size 1–7 and one of three aggregation rules (blue). Attacks (red) exercise the four adaptive threat classes; baselines (blue, below) provide the comparison points. Outputs are the security and cost metrics (purple) collected into the result tables and plots. All reported numbers are placeholders until measured.

RQ2 (second-order injection). Does payload isolation reduce the verdict shift ε (Definition 1) induced by JudgeDeceiver-style injection, and does it lower ASR-UC when injection is combined with judge compromise?

RQ3 (adaptivity). How much does an attacker that optimises against the specific aggregation rule increase ASR-UC relative to a static attacker, for each of cmed, gmed, Krum, and majority vote?

RQ4 (correlated failure). How does the homogeneous-vs-diverse backbone choice affect the honest-error correlation ρ and, through it, ASR-UC and ORR?

RQ5 (cost). What is the security-versus-cost frontier (Table 4) in measured latency and token cost as n grows from 1 to 7?

RQ6 (utility). What is the over-refusal rate on benign traffic, and how much benign utility is retained relative to no defense?

10.2. Datasets and payloads

Adversarial payloads are drawn from established jailbreak and prompt-injection resources: in-the-wild jailbreak prompts [2], automated jailbreaks

from GCG/AdvBench [1], PAIR [3], TAP [62], and GPTFuzzer [63], evaluated within the HarmBench harness [64]; prompt-injection and task-integrity payloads from the formal injection benchmark [6], InjecAgent [65], and universal injection [66]. Benign traffic (for ORR and utility) is drawn from instruction-following and QA distributions disjoint from the attack sets. Second-order injections against the judges are generated with JudgeDeceiver [27]. Ground-truth labels y^* follow each benchmark’s released annotations; task-integrity labels use the instruction-hierarchy criterion of [7].

10.3. Models and configurations

Judges and analyzers are instantiated on open backbones (Llama-3, Qwen2.5, Mistral) and proprietary backbones (GPT-4o, Claude-3.5), in *homogeneous* committees (all judges share a backbone) and *diverse* committees (backbones and prompts varied). Judges are hardened by SecAlign-style preference optimisation [8] and StruQ-style structured-input training [9]; scores are temperature-calibrated [55]; rationale embeddings use a fixed sentence encoder [56]. Committee size sweeps $n \in \{1, \dots, 7\}$.

10.4. Baselines

1. **AutoDefense** [14]: multi-agent filter with a single Coordinator (no Byzantine tolerance, no payload isolation).
2. **Single hardened model** [8]: one SecAlign judge (the “component-3-alone” baseline of Section 6.5).
3. **Plain majority vote**: committee with decision-only aggregation (Section 3).

Secondary reference points for context, not head-to-head superiority claims, include single-model guardrails [40, 43] and content moderators [10, 13].

10.5. Attacks (adaptive by construction)

1. **Compromised judge**: f judges return adversary-chosen verdict vectors.
2. **Colluding defenders**: the f Byzantine judges coordinate to stay within the honest spread while biasing the aggregate, mirroring “a-little-is-enough” [31].
3. **Second-order payload injection**: JudgeDeceiver [27] embeds instructions in the payload to shift honest verdicts.

4. **Adaptive-on-aggregation:** the attacker optimises against the specific rule, as inner-product manipulation [32] and local model poisoning [30] do against Krum/median.

The colluding fraction is swept from independent single compromise to the boundary $2f+2 < n$; each attack is run in static and adaptive variants.

10.6. Metrics

Defense success rate; ASR-UC (3) and targeted-ASR; over-refusal rate ORR (4); Byzantine tolerance fraction; detection F1/AUROC of malicious verdicts (treating aggregation as an outlier detector); latency (Eq. (10)); token/API cost (Eq. (9)); and benign utility retention. We additionally report the measured honest radius r , margin γ , and correlation ρ , since Theorem 1 and Proposition 3 predict behaviour through exactly these quantities.

10.7. Ablations

Robust aggregation on/off (vs. single coordinator); payload isolation on/off; hardened judges on/off; committee size $n \in \{1, \dots, 7\}$; colluding fraction; and homogeneous vs. diverse backbones (the correlated-failure test for RQ4). Each ablation isolates one of the four components of Section 6.

10.8. Reproducibility checklist

Fixed random seeds and per-request isolation nonces logged; exact backbone versions/checkpoints and decoding parameters recorded; attack budgets (queries, iterations) fixed per benchmark; aggregation tolerances (η for gmed, assumed f for Krum) reported; all placeholder cells replaced with mean $\pm$ standard deviation over seeds and, where comparisons are drawn, an explicit significance test; released code and payload manifests. No result is to be reported without its measurement conditions.

10.9. Placeholder result templates

Tables 5 and 6 and Fig. 6 give the shape of the intended reporting. Every cell reads “—” and every curve is labelled a conceptual expected trend.

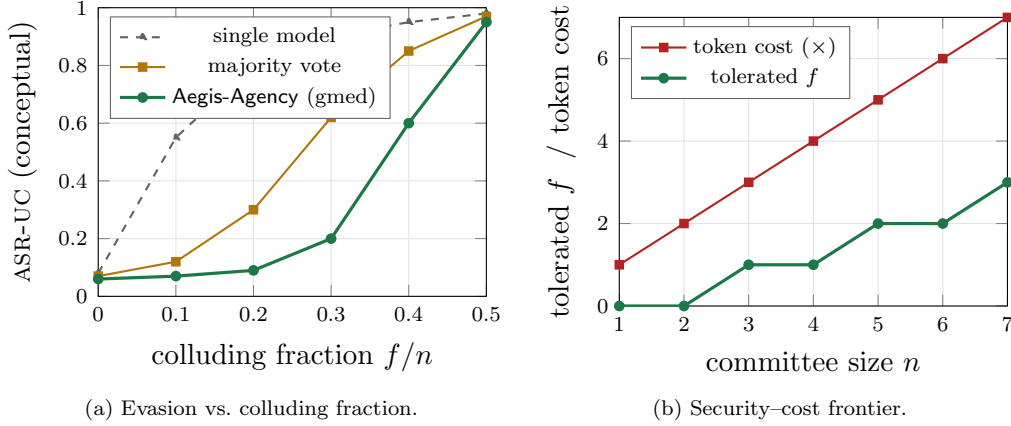


Figure 6: **Conceptual expected trends; not experimental results.** *Placeholder curves; replace with actual experimental data before submission.* (a) Expected evasion (ASR-UC) as the colluding fraction grows: a single model fails as soon as it is compromised, majority vote degrades earlier, and robust aggregation is expected to stay low until the tolerance boundary $f < n/2$ before rising (graceful degradation, G3). (b) The security-cost frontier of Table 4: token cost grows linearly in n while tolerated faults grow as $\lfloor (n-1)/2 \rfloor$. Shapes are illustrative, contain no measured data, and encode the hypotheses the evaluation will test.

11. Discussion

What the guarantees do and do not deliver.. Theorem 1 shows that robust verdict aggregation converts a Byzantine minority into a bounded, and under a sufficient margin harmless, perturbation of the decision. The force of the result is entirely in its condition $\gamma > C_\alpha r$: it moves the burden of proof from “is the pipeline secure?” to the measurable “do honest judges concentrate tightly relative to their distance from the policy boundary?” This is progress precisely because it is falsifiable— r , γ , and ρ are quantities Section 10 measures, and a deployment in which honest judges are diffuse or the margin is small will show it in those numbers rather than in a surprise failure.

Why aggregation in the verdict space, not over decisions.. Plain majority vote is the natural committee baseline, but it aggregates only d_k and is therefore blind to a colluder that votes with the honest majority while manipulating scores or rationales—to bias a downstream threshold, to poison an audit trail, or to prepare a later escalation. Aggregating $(s_k, e_k) \in \mathbb{R}^d$ makes such a colluder a geometric outlier that gmed and KRUM down-weight or reject. The cost is that the verdict space must be well-behaved: the rationale embedding

Table 5: *Placeholder / expected-trend template; replace with measured results before submission.* Main comparison at fixed diverse-backbone committee. Lower ASR-UC and ORR are better. No numbers are filled because none have been measured.

Method	n	ASR-UC($f=1$)	ASR-UC($f=2$)	ORR	Tokens ($\times$)
Single hardened model [8]	1	—	n/a	—	1
AutoDefense (single coord.) [14]	3	—	—	—	3
Plain majority vote	5	—	—	—	5
Aegis-Agency (cmed)	5	—	—	—	5
Aegis-Agency (gmed)	5	—	—	—	5
Aegis-Agency (KRUM)	5	—	—	—	5

Table 6: *Placeholder / expected-trend template; replace with measured results before submission.* Ablation over the four components; each row disables one mechanism relative to full Aegis-Agency. Cells are intentionally empty.

Configuration	Robust agg.	Isolation	Hardened judges	ASR-UC($f=1$)
Full Aegis-Agency	✓	✓	✓	—
— robust aggregation (single coord.)		✓	✓	—
— payload isolation	✓		✓	—
— hardened judges	✓	✓		—
— all (AutoDefense-like)				—

must place honest justifications near one another and manipulated ones far away, which is an assumption on the encoder that we make explicit and test, not one we take for granted.

Connection to robust learning, and its warning.. We deliberately import machinery whose limits are documented. Adaptive attacks defeat Krum and median in federated learning when the attacker stays within the honest variance [31, 30, 32], and single-selection rules degrade in high dimension [46]. Both warnings transfer. The first is why our evaluation is adaptive by construction and why Theorem 1 is stated as a conditional guarantee rather than an unconditional one; a colluding committee that keeps its verdicts within the honest radius r does not violate the theorem but does enlarge the effective r , and that enlargement is exactly what the colluding-fraction sweep measures. The second is why we keep the verdict space low-dimensional and treat KRUM as one option among three.

Diversity is a requirement, not a tuning knob.. Proposition 3 is the least comfortable result in the paper and the most important: once honest errors are correlated, adding judges stops helping, and in the limit of identical failure the committee is a single point of failure with a larger bill. Backbone and prompt diversity is the only lever that reduces ρ , and the ensemble literature cautions that diversity does not translate cleanly into accuracy [38]. We therefore report ρ alongside every robustness number and treat the homogeneous-vs-diverse gap as a headline measurement rather than an ablation footnote.

Relation to the broader multi-agent security agenda.. Recent work shows task-performing agent teams are attackable through their interactions [21, 24, 23, 44, 25, 26]. **Aegis-Agency** addresses the mirror-image problem for a *defensive* team and, by using a star topology with no inter-agent messages, removes the communication channel those attacks exploit—at the cost of forgoing the benefits of agent debate [19]. Whether a debating defense committee could achieve tighter honest concentration (smaller r) without reopening the communication surface is an open question our architecture is positioned to study but does not resolve.

12. Limitations

We state the limitations plainly; several are structural and bound what the approach can claim.

Conditional, not unconditional, guarantees.. Theorem 1 holds only when honest judges concentrate (r small) and the decision margin is large ($\gamma > C_\alpha r$). Neither is guaranteed in deployment: ambiguous payloads, shared systematic biases [28, 29], and near-boundary policies can violate both. The theorem does not certify security; it converts security into measurable preconditions.

Small committees tolerate few faults.. With $n \in \{3, 4, 5\}$ the tolerance is $f \in \{1, 1, 2\}$ (Table 4). Whether tolerating a single compromised judge is meaningful robustness for a given deployment is an empirical and threat-model question, not one the theory settles. Large committees are the natural answer but multiply token and latency cost linearly (Section 9).

Correlated failure is mitigated, not solved.. Proposition 3 shows that once honest errors are correlated, added judges stop helping; diversity reduces but does not eliminate ρ , and the required diversity may be hard to achieve when strong judges cluster on a few backbones. The limiting case of identical failure across all judges is explicitly out of scope.

Adaptive attacks may enlarge the effective radius.. A colluding committee that keeps its verdicts within the honest spread does not break Theorem 1 but enlarges the effective r , and history shows adaptive attackers find such strategies against robust aggregators [31, 30, 32]. Our evaluation is adaptive, but no finite adaptive suite proves robustness against all future attacks.

Isolation leakage is empirical.. Proposition 2 is conditional on the measured leakage ε ; perfect isolation is an idealisation. A judge whose fine-tuning is imperfect will have $\varepsilon > 0$, reopening a fraction of the second-order surface.

Embedding-space assumptions.. Aggregating rationale embeddings assumes the encoder separates honest from manipulated justifications; a colluder that produces honest-looking rationales for a dishonest verdict stresses this assumption, and encoder robustness is itself an attack surface we do not fully characterise.

Placeholder evaluation.. The reported tables and plots are protocol templates, not measurements. Every quantitative claim about **Aegis-Agency**’s effectiveness is therefore pending the experiments of Section 10; the contribution at this stage is the threat model, architecture, and analysis, together with a falsifiable evaluation plan.

13. Conclusion

Multi-agent pipelines that filter LLM outputs have made the content they inspect safer while leaving themselves unexamined. We argued that such a pipeline is a multi-agent system whose own security matters, and we studied two threats with no single-model analogue: compromised or colluding internal defenders, and second-order prompt injection carried by the content the judges read. **Aegis-Agency** answers both by replacing the single trusted coordinator with Byzantine-robust aggregation of judge verdicts, isolating the untrusted payload from the judges’ instruction channel, hardening each judge, and generalising the verdict target from jailbreak to task-integrity.

We proved, under stated honest-concentration and margin assumptions, that robust aggregation prevents a Byzantine minority ($2f+2 < n$ for selection, $f < n/2$ for median rules) from flipping the aggregate decision in either direction; we characterised how the guarantee degrades once honest failures correlate, turning backbone diversity from a tuning choice into a requirement; and we quantified the linear token/latency cost the committee incurs. We were explicit throughout that a lower attack success rate under compromise is evidence of graceful degradation, not a proof of security, and that the empirical figures here are placeholders fixing a falsifiable protocol.

Two directions follow directly. First, executing the adaptive evaluation of Section 10—especially the homogeneous-vs-diverse correlated-failure test and the security-versus-cost frontier—will tell us whether the preconditions of Theorem 1 hold for real hardened LLM judges, which is the crux of whether the approach is practically meaningful at small n . Second, tightening honest concentration without reopening the inter-agent communication surface—for instance through structured, non-adversarial consensus among judges—could relax the margin condition and extend tolerance, and is a question the star-topology architecture is positioned to study. We hope the framing of the defense pipeline as a first-class attack surface, and the insistence on stating exactly where robustness stops, are useful beyond the specific architecture proposed here.

Appendix A. Proofs

This appendix gives the complete proofs of the results marked complete in Section 8 and the deferred specialisations of the sketched results.

Appendix A.1. Proof of Lemma 1 (geometric-median displacement)

Proof. Let $\hat{\mathbf{u}} = \arg \min_{\mathbf{y} \in \mathbb{R}^d} F(\mathbf{y})$ with $F(\mathbf{y}) = \sum_{k=1}^n \|\mathbf{y} - \mathbf{u}_k\|$, the geometric median. Optimality gives $F(\hat{\mathbf{u}}) \leq F(\mathbf{u}^*)$, i.e.

$$\sum_{k \in \mathcal{H}} \|\hat{\mathbf{u}} - \mathbf{u}_k\| + \sum_{k \in \mathcal{B}} \|\hat{\mathbf{u}} - \mathbf{u}_k\| \leq \sum_{k \in \mathcal{H}} \|\mathbf{u}^* - \mathbf{u}_k\| + \sum_{k \in \mathcal{B}} \|\mathbf{u}^* - \mathbf{u}_k\|. \quad (\text{A.1})$$

Rearranging Eq. (A.1),

$$\sum_{k \in \mathcal{H}} (\|\hat{\mathbf{u}} - \mathbf{u}_k\| - \|\mathbf{u}^* - \mathbf{u}_k\|) \leq \sum_{k \in \mathcal{B}} (\|\mathbf{u}^* - \mathbf{u}_k\| - \|\hat{\mathbf{u}} - \mathbf{u}_k\|). \quad (\text{A.2})$$

For the right-hand side, the reverse triangle inequality gives, for every k , $\|\mathbf{u}^* - \mathbf{u}_k\| - \|\hat{\mathbf{u}} - \mathbf{u}_k\| \leq \|\mathbf{u}^* - \hat{\mathbf{u}}\|$, hence

$$\text{RHS of (A.2)} \leq |\mathcal{B}| \|\mathbf{u}^* - \hat{\mathbf{u}}\| = f \|\hat{\mathbf{u}} - \mathbf{u}^*\|. \quad (\text{A.3})$$

For the left-hand side, fix an honest $k \in \mathcal{H}$. By the triangle inequality $\|\hat{\mathbf{u}} - \mathbf{u}_k\| \geq \|\hat{\mathbf{u}} - \mathbf{u}^*\| - \|\mathbf{u}^* - \mathbf{u}_k\|$, and since $\|\mathbf{u}^* - \mathbf{u}_k\| \leq r$ (Assumption 1),

$$\|\hat{\mathbf{u}} - \mathbf{u}_k\| - \|\mathbf{u}^* - \mathbf{u}_k\| \geq \|\hat{\mathbf{u}} - \mathbf{u}^*\| - \|\mathbf{u}^* - \mathbf{u}_k\| - \|\mathbf{u}^* - \mathbf{u}_k\| \geq \|\hat{\mathbf{u}} - \mathbf{u}^*\| - 2r. \quad (\text{A.4})$$

Summing Eq. (A.4) over the $|\mathcal{H}| = n - f$ honest judges,

$$\text{LHS of (A.2)} \geq (n - f)(\|\hat{\mathbf{u}} - \mathbf{u}^*\| - 2r). \quad (\text{A.5})$$

Combining Eqs. (A.3) and (A.5) with Eq. (A.2),

$$(n - f)(\|\hat{\mathbf{u}} - \mathbf{u}^*\| - 2r) \leq f \|\hat{\mathbf{u}} - \mathbf{u}^*\| \implies (n - 2f) \|\hat{\mathbf{u}} - \mathbf{u}^*\| \leq 2(n - f)r. \quad (\text{A.6})$$

Since $f < n/2$ we have $n - 2f > 0$, so dividing yields $\|\hat{\mathbf{u}} - \mathbf{u}^*\| \leq \frac{2(n-f)}{n-2f} r = C_\alpha r$ with $\alpha = f/n$. The Byzantine verdicts entered only through Eq. (A.3), where the bound is independent of their values, so the displacement bound holds for arbitrary Byzantine $\{\mathbf{u}_k\}_{k \in \mathcal{B}}$. $\square$

Remark 6. The constant $C_\alpha = \frac{2(1-\alpha)}{1-2\alpha}$ is a *sufficient* displacement bound, consistent with the breakdown point $\frac{1}{2}$ of the geometric median [47, 48]. We make no optimality or matching claim: a smaller constant may hold under additional assumptions, and the divergence as $\alpha \rightarrow \frac{1}{2}$ is intrinsic to any minority-tolerance estimator.

Appendix A.2. Proof of Theorem 1 (decision integrity)

Proof. By Lemma 1 (for gmed) the aggregate satisfies $\|\hat{\mathbf{u}} - \mathbf{u}^*\| \leq C_\alpha r$. The score coordinate is a 1-Lipschitz projection, $|\pi_1(\hat{\mathbf{u}}) - \pi_1(\mathbf{u}^*)| \leq \|\hat{\mathbf{u}} - \mathbf{u}^*\| \leq C_\alpha r$. By Assumption 2, $\pi_1(\mathbf{u}^*)$ lies at distance at least γ from the threshold τ . If $\gamma > C_\alpha r$ (Eq. (7)), then $\pi_1(\hat{\mathbf{u}})$ stays on the same side of τ as $\pi_1(\mathbf{u}^*)$:

$$\pi_1(\mathbf{u}^*) \geq \tau + \gamma \implies \pi_1(\hat{\mathbf{u}}) \geq \tau + \gamma - C_\alpha r > \tau, \quad \pi_1(\mathbf{u}^*) \leq \tau - \gamma \implies \pi_1(\hat{\mathbf{u}}) \leq \tau - \gamma + C_\alpha r < \tau. \quad (\text{A.7})$$

Hence $D(\hat{\mathbf{u}}) = \mathbf{1}\{\pi_1(\hat{\mathbf{u}}) \geq \tau\} = \mathbf{1}\{\pi_1(\mathbf{u}^*) \geq \tau\} = D(\mathbf{u}^*) = y^*(x)$, for every choice of the Byzantine verdicts. Both directions are covered: the payload the honest committee would block is blocked (no forced pass), and the one it would allow is allowed (no forced block). $\square$

Proof of Corollary 1. The two implications in the proof of Theorem 1 are symmetric in the sign of $\pi_1(\mathbf{u}^\star) - \tau$; the second implication is exactly the statement that a benign payload ($\pi_1(\mathbf{u}^\star) \leq \tau - \gamma$, so $\mathbf{D}(\mathbf{u}^\star) = 0$) cannot be pushed to $\hat{d} = 1$ by the Byzantine judges. $\square$

Appendix A.3. Proof of Proposition 1 (coordinate-wise median)

Proof. Fix a coordinate j and consider the n scalars $\{\pi_j(\mathbf{u}_k)\}_{k \in [n]}$, of which at most f are Byzantine. Let $a = \min_{k \in \mathcal{H}} \pi_j(\mathbf{u}_k)$ and $b = \max_{k \in \mathcal{H}} \pi_j(\mathbf{u}_k)$ be the honest range. Among all n values, at least $n - f > n/2$ lie in $[a, b]$. A median (any order statistic of rank $\lceil n/2 \rceil$ or $\lfloor n/2 \rfloor + 1$) of n numbers of which strictly more than half lie in $[a, b]$ must itself lie in $[a, b]$: fewer than $n/2$ values lie strictly below a and fewer than $n/2$ strictly above b , so the middle order statistic is bracketed. Thus $\pi_j(\hat{\mathbf{u}}) \in [a, b]$. For $j = 1$ (score), $a, b \in [\pi_1(\mathbf{u}^\star) - r, \pi_1(\mathbf{u}^\star) + r]$ by Assumption 1, giving $|\pi_1(\hat{\mathbf{u}}) - \pi_1(\mathbf{u}^\star)| \leq r$. Substituting the displacement r (i.e. $C_\alpha = 1$) into the margin argument of Theorem 1 yields the decision guarantee whenever $\gamma > r$. $\square$

Appendix A.4. Proof of Proposition 2 (isolation containment)

Proof. Fix an honest judge k with verdict law $P_k(\cdot \mid c, x)$. Its decision is the event $A = \{v : \mathbf{D} = 1\}$. By ε -isolation (Definition 1), $|P_k(A \mid c, x) - P_k(A \mid c, x')| \leq \text{TV}(P_k(\cdot \mid c, x), P_k(\cdot \mid c, x')) \leq \varepsilon$, so the injection changes judge k 's decision probability by at most ε ; the probability it flips a fixed drawn decision is at most ε . If honest judges draw independently, the number of flipped honest decisions is stochastically dominated by a Binomial($n - f, \varepsilon$), whence $\mathbb{P}[\text{more than } t \text{ flips}] \leq \binom{n-f}{t+1} \varepsilon^{t+1}$ by the union bound over $(t+1)$ -subsets. When $\varepsilon = 0$ every honest verdict law is invariant to the injection, so no honest verdict changes and the attacker must instead corrupt judges directly, which is a Byzantine action counted in f and governed by Theorem 1. $\square$

Appendix A.5. Proof of Proposition 3 (correlated honest failure)

Proof. Let $\{Z_k\}_{k \in \mathcal{H}}$ be exchangeable Bernoulli(μ) with $\text{Corr}(Z_i, Z_j) = \rho$ for $i \neq j$, so $\text{Cov}(Z_i, Z_j) = \rho \mu(1 - \mu)$. With $N = n - f$ and $\bar{Z} = \frac{1}{N} \sum_k Z_k$,

$$\text{Var}(\bar{Z}) = \frac{1}{N^2} \left(\sum_k \text{Var}(Z_k) + \sum_{i \neq j} \text{Cov}(Z_i, Z_j) \right) = \frac{1}{N^2} \left(N\mu(1-\mu) + N(N-1)\rho\mu(1-\mu) \right), \quad (\text{A.8})$$

which simplifies to $\text{Var}(\bar{Z}) = \mu(1-\mu) \left[\frac{1-\rho}{N} + \rho \right]$, proving Eq. (8). As $N \rightarrow \infty$ the first term vanishes and $\text{Var}(\bar{Z}) \rightarrow \mu(1-\mu)\rho$, which is strictly positive for $\rho \in (0, 1]$ and $\mu \in (0, 1)$; the sample mean does not concentrate at μ . For the majority-failure lower bound, a correct honest majority requires $\bar{Z} > \frac{1}{2}$. If $\mu \leq \frac{1}{2}$ then $\{\bar{Z} > \frac{1}{2}\} \subseteq \{\bar{Z} - \mu > \frac{1}{2} - \mu\}$ and, when $\mu < \frac{1}{2}$, Chebyshev gives $\mathbb{P}[\bar{Z} > \frac{1}{2}] \leq \mathbb{P}[|\bar{Z} - \mu| \geq \frac{1}{2} - \mu] \leq \frac{\text{Var}(\bar{Z})}{(\frac{1}{2} - \mu)^2}$. As $N \rightarrow \infty$ this upper bound tends to $\frac{\mu(1-\mu)\rho}{(\frac{1}{2} - \mu)^2}$, a constant independent of N ; hence for $\rho > 0$ the honest majority cannot be made correct with probability approaching one merely by enlarging the committee. Reducing ρ (through backbone/prompt diversity) is the only term in Eq. (8) that restores $1/N$ -style concentration. $\square$

Remark 7. The bound is one-sided and asymptotic in N ; it establishes that correlation imposes a ρ -dependent floor on committee variance, which is the formal content of the idea’s “vacuous if all judges fail identically.” It is a limiting-case impossibility for the honest majority, not a defense, and we present it as such.

Appendix B. Additional Experimental Detail

This appendix expands the protocol of Section 10. As there, **all figures and tables in this appendix are placeholders / expected trends and must be replaced with measured results before submission.**

Appendix B.1. Attacker instantiation

For the *colluding-defenders* attack we adopt the “a-little-is-enough” principle [31]: the f Byzantine judges each report a verdict vector $\mathbf{u}_k = \mathbf{u}^* + z \hat{\mathbf{g}}$, where $\hat{\mathbf{g}}$ is the estimated direction toward the decision boundary and $z \leq r$ keeps each Byzantine verdict inside the honest radius so that robust aggregation cannot flag it as an outlier. The colluding-fraction sweep increases f until $2f+2 < n$ is violated. For the *adaptive-on-aggregation* attack we optimise $\{\mathbf{u}_k\}_{k \in \mathcal{B}}$ against the specific rule: inner-product manipulation [32] for gmed/cmed and the local-model-poisoning objective [30] for KRUM. The *second-order injection* attack uses JudgeDeceiver [27] to craft payload suffixes, evaluated with isolation on and off to estimate ε (Definition 1).

Appendix B.2. Measuring the theory’s quantities

Theorem 1 and Proposition 3 predict behaviour through r , γ , and ρ ; we measure all three. The honest radius r is the empirical 95th-percentile of $\|\mathbf{u}_k - \mathbf{u}^*\|$ over honest judges with $\mathbf{u}^*$ the honest geometric median. The margin γ is $|\pi_1(\mathbf{u}^*) - \tau|$ averaged over payloads, reported separately for benign and adversarial classes. The correlation ρ is the mean pairwise correlation of honest correctness indicators, reported for homogeneous and diverse committees to test RQ4. Plotting ASR-UC against the measured $C_\alpha r/\gamma$ ratio provides a direct check of Eq. (7): the theory predicts a rise in ASR-UC as $C_\alpha r/\gamma$ crosses 1.

Appendix B.3. Placeholder frontier and sweeps

Figure 6 (main text) shows the two conceptual sweeps: ASR-UC versus colluding fraction, and the security–cost frontier versus n . Both are drawn as expected trends with no data. The correlated-failure ablation is reported as paired (homogeneous, diverse) points at each n , with ρ annotated.

Appendix B.4. Reproducibility artefacts

On release we provide: backbone checkpoints/versions and decoding parameters; the isolation delimiter scheme and per-request nonce generation; hardening (SecAlign/StruQ) training configurations; aggregation tolerances (η , assumed f); attack budgets per benchmark; seeds; and the payload manifests mapping each item to its source benchmark and ground-truth label. Each released number carries its measurement conditions and dispersion across seeds.

References

- [1] A. Zou, Z. Wang, N. Carlini, M. Nasr, J. Z. Kolter, M. Fredrikson, Universal and transferable adversarial attacks on aligned language models, arXiv preprint arXiv:2307.15043 (2023). [arXiv:2307.15043](#).
- [2] X. Shen, Z. Chen, M. Backes, Y. Shen, Y. Zhang, "Do Anything Now": Characterizing and evaluating in-the-wild jailbreak prompts on large language models, in: Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS), 2024. [arXiv:2308.03825](#).

- [3] P. Chao, A. Robey, E. Dobriban, H. Hassani, G. J. Pappas, E. Wong, Jailbreaking black box large language models in twenty queries, arXiv preprint arXiv:2310.08419 (2023). **arXiv:2310.08419**.
- [4] S. Yi, Y. Liu, Z. Sun, T. Cong, X. He, J. Song, K. Xu, Q. Li, Jailbreak attacks and defenses against large language models: A survey, arXiv preprint arXiv:2407.04295 (2024). **arXiv:2407.04295**.
- [5] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, M. Fritz, Not what you’ve signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection, in: Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec), 2023. **arXiv:2302.12173**.
- [6] Y. Liu, Y. Jia, R. Geng, J. Jia, N. Z. Gong, Formalizing and benchmarking prompt injection attacks and defenses, in: 33rd USENIX Security Symposium (USENIX Security 24), 2024. **arXiv:2310.12815**.
- [7] E. Wallace, K. Xiao, R. Leike, L. Weng, J. Heidecke, A. Beutel, The instruction hierarchy: Training LLMs to prioritize privileged instructions, arXiv preprint arXiv:2404.13208 (2024). **arXiv:2404.13208**.
- [8] S. Chen, A. Zharmagambetov, S. Mahloujifar, K. Chaudhuri, D. Wagner, C. Guo, SecAlign: Defending against prompt injection with preference optimization, in: Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security (CCS), 2025. **arXiv:2410.05451**.
- [9] S. Chen, J. Piet, C. Sitawarin, D. Wagner, StruQ: Defending against prompt injection with structured queries, in: 34th USENIX Security Symposium (USENIX Security 25), 2025. **arXiv:2402.06363**.
- [10] H. Inan, K. Upasani, J. Chi, R. Rungta, K. Iyer, Y. Mao, M. Tontchev, Q. Hu, B. Fuller, D. Testuggine, M. Khabsa, Llama Guard: LLM-based input-output safeguard for human-AI conversations, arXiv preprint arXiv:2312.06674 (2023). **arXiv:2312.06674**.
- [11] T. Rebedea, R. Dinu, M. Sreedhar, C. Parisien, J. Cohen, NeMo Guardrails: A toolkit for controllable and safe LLM applications with

- programmable rails, in: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP): System Demonstrations, 2023. [arXiv:2310.10501](#).
- [12] T. Markov, C. Zhang, S. Agarwal, T. Eloundou, T. Lee, S. Adler, A. Jiang, L. Weng, A holistic approach to undesired content detection in the real world, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2023. [arXiv:2208.03274](#).
 - [13] W. Zeng, Y. Liu, R. Mullins, L. Peran, J. Fernandez, H. Harkous, K. Narasimhan, D. Proud, P. Kumar, B. Radharapu, O. Sturman, O. Wahltinez, ShieldGemma: Generative AI content moderation based on Gemma, arXiv preprint [arXiv:2407.21772](#) (2024). [arXiv:2407.21772](#).
 - [14] Y. Zeng, Y. Wu, X. Zhang, H. Wang, Q. Wu, AutoDefense: Multi-agent LLM defense against jailbreak attacks, arXiv preprint [arXiv:2403.04783](#) (2024). [arXiv:2403.04783](#).
 - [15] Q. Wu, G. Bansal, J. Zhang, Y. Wu, B. Li, E. Zhu, L. Jiang, X. Zhang, S. Zhang, J. Liu, A. H. Awadallah, R. W. White, D. Burger, C. Wang, AutoGen: Enabling next-gen LLM applications via multi-agent conversation, arXiv preprint [arXiv:2308.08155](#) (2023). [arXiv:2308.08155](#).
 - [16] S. Hong, M. Zhuge, J. Chen, X. Zheng, Y. Cheng, C. Zhang, J. Wang, Z. Wang, S. K. S. Yau, Z. Lin, L. Zhou, C. Ran, L. Xiao, C. Wu, J. Schmidhuber, MetaGPT: Meta programming for a multi-agent collaborative framework, in: International Conference on Learning Representations (ICLR), 2024. [arXiv:2308.00352](#).
 - [17] G. Li, H. A. A. K. Hammoud, H. Itani, D. Khizbullin, B. Ghanem, CAMEL: Communicative agents for “mind” exploration of large language model society, in: Advances in Neural Information Processing Systems (NeurIPS) 36, 2023. [arXiv:2303.17760](#).
 - [18] C. Qian, W. Liu, H. Liu, N. Chen, Y. Dang, J. Li, C. Yang, W. Chen, Y. Su, X. Cong, J. Xu, D. Li, Z. Liu, M. Sun, ChatDev: Communicative agents for software development, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL), 2024. [arXiv:2307.07924](#).

- [19] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, I. Mordatch, Improving factuality and reasoning in language models through multiagent debate, in: Proceedings of the 41st International Conference on Machine Learning (ICML), 2024. [arXiv:2305.14325](#).
- [20] Z. Xi, W. Chen, X. Guo, W. He, Y. Ding, B. Hong, M. Zhang, J. Wang, S. Jin, E. Zhou, et al., The rise and potential of large language model based agents: A survey, arXiv preprint [arXiv:2309.07864](#) (2023). [arXiv:2309.07864](#).
- [21] D. Lee, M. Tiwari, Prompt infection: LLM-to-LLM prompt injection within multi-agent systems, arXiv preprint [arXiv:2410.07283](#) (2024). [arXiv:2410.07283](#).
- [22] Y. Tian, X. Yang, J. Zhang, Y. Dong, H. Su, Evil geniuses: Delving into the safety of LLM-based agents, arXiv preprint [arXiv:2311.11855](#) (2023). [arXiv:2311.11855](#).
- [23] T. Ju, Y. Wang, X. Ma, P. Cheng, H. Zhao, Y. Wang, L. Liu, J. Xie, Z. Zhang, G. Liu, Flooding spread of manipulated knowledge in LLM-based multi-agent communities, arXiv preprint [arXiv:2407.07791](#) (2024). [arXiv:2407.07791](#).
- [24] P. He, Y. Lin, S. Dong, H. Xu, Y. Xing, H. Liu, Red-teaming LLM multi-agent systems via communication attacks, in: Findings of the Association for Computational Linguistics (ACL Findings), 2025. [arXiv:2502.14847](#).
- [25] Z. Zhang, Y. Zhang, L. Li, H. Gao, L. Wang, H. Lu, F. Zhao, Y. Qiao, J. Shao, PsySafe: A comprehensive framework for psychological-based attack, defense, and evaluation of multi-agent system safety, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL), 2024. [arXiv:2401.11880](#).
- [26] H. Zhang, J. Huang, K. Mei, Y. Yao, Z. Wang, C. Zhan, H. Wang, Y. Zhang, Agent security bench (ASB): Formalizing and benchmarking attacks and defenses in LLM-based agents, in: International Conference on Learning Representations (ICLR), 2025. [arXiv:2410.02644](#).

- [27] J. Shi, Z. Yuan, Y. Liu, Y. Huang, P. Zhou, L. Sun, N. Z. Gong, Optimization-based prompt injection attack to LLM-as-a-judge, in: Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS), 2024. [arXiv:2403.17710](#).
- [28] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, I. Stoica, Judging LLM-as-a-judge with MT-Bench and Chatbot Arena, in: Advances in Neural Information Processing Systems (NeurIPS) 36, Datasets and Benchmarks Track, 2023. [arXiv:2306.05685](#).
- [29] P. Wang, L. Li, L. Chen, Z. Cai, D. Zhu, B. Lin, Y. Cao, Q. Liu, T. Liu, Z. Sui, Large language models are not fair evaluators, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL), 2024. [arXiv:2305.17926](#).
- [30] M. Fang, X. Cao, J. Jia, N. Z. Gong, Local model poisoning attacks to Byzantine-robust federated learning, in: 29th USENIX Security Symposium (USENIX Security 20), 2020, pp. 1605–1622. [arXiv:1911.11815](#).
- [31] G. Baruch, M. Baruch, Y. Goldberg, A little is enough: Circumventing defenses for distributed learning, in: Advances in Neural Information Processing Systems (NeurIPS) 32, 2019. [arXiv:1902.06156](#).
- [32] C. Xie, O. Koyejo, I. Gupta, Fall of empires: Breaking Byzantine-tolerant SGD by inner product manipulation, in: Proceedings of the 35th Conference on Uncertainty in Artificial Intelligence (UAI), PMLR 115, 2019, pp. 261–270. [arXiv:1903.03936](#).
- [33] P. Blanchard, E. M. El Mhamdi, R. Guerraoui, J. Stainer, Machine learning with adversaries: Byzantine tolerant gradient descent, in: Advances in Neural Information Processing Systems (NeurIPS) 30, 2017.
- [34] D. Yin, Y. Chen, K. Ramchandran, P. Bartlett, Byzantine-robust distributed learning: Towards optimal statistical rates, in: Proceedings of the 35th International Conference on Machine Learning (ICML), PMLR 80, 2018, pp. 5650–5659. [arXiv:1803.01498](#).
- [35] Y. Chen, L. Su, J. Xu, Distributed statistical machine learning in adversarial settings: Byzantine gradient descent, Proceedings of the ACM

- on Measurement and Analysis of Computing Systems (POMACS) 1 (2) (2017) 1–25. doi:10.1145/3154503.
- [36] K. Pillutla, S. M. Kakade, Z. Harchaoui, Robust aggregation for federated learning, *IEEE Transactions on Signal Processing* 70 (2022) 1142–1154. doi:10.1109/TSP.2022.3153135.
 - [37] K. Hines, G. Lopez, M. Hall, F. Zarfati, Y. Zunger, E. Kiciman, Defending against indirect prompt injection attacks with spotlighting, *arXiv preprint arXiv:2403.14720* (2024). arXiv:2403.14720.
 - [38] L. I. Kuncheva, C. J. Whitaker, Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy, *Machine Learning* 51 (2) (2003) 181–207. doi:10.1023/A:1022859003006.
 - [39] S. Chennabasappa, C. Nikolaidis, D. Song, D. Molnar, S. Ding, S. Wan, S. Whitman, L. Deason, N. Doucette, A. Montilla, A. Gampa, B. de Paola, D. Gabi, J. Crnkovich, J.-C. Testud, K. He, R. Chaturvedi, W. Zhou, J. Saxe, LlamaFirewall: An open source guardrail system for building secure AI agents, *arXiv preprint arXiv:2505.03574* (2025). arXiv:2505.03574.
 - [40] A. Robey, E. Wong, H. Hassani, G. J. Pappas, SmoothLLM: Defending large language models against jailbreaking attacks, *arXiv preprint arXiv:2310.03684* (2023). arXiv:2310.03684.
 - [41] R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, C. Finn, Direct preference optimization: Your language model is secretly a reward model, in: *Advances in Neural Information Processing Systems (NeurIPS)* 36, 2023. arXiv:2305.18290.
 - [42] A. Kumar, C. Agarwal, S. Srinivas, A. J. Li, S. Feizi, H. Lakkaraju, Certifying LLM safety against adversarial prompting, in: *Conference on Language Modeling (COLM)*, 2024. arXiv:2309.02705.
 - [43] N. Jain, A. Schwarzschild, Y. Wen, G. Somepalli, J. Kirchenbauer, P.-y. Chiang, M. Goldblum, A. Saha, J. Geiping, T. Goldstein, Baseline defenses for adversarial attacks against aligned language models, *arXiv preprint arXiv:2309.00614* (2023). arXiv:2309.00614.

- [44] Z. Zhou, Z. Li, J. Zhang, Y. Zhang, K. Wang, Y. Liu, Q. Guo, CORBA: Contagious recursive blocking attacks on multi-agent systems based on large language models, arXiv preprint arXiv:2502.14529 (2025). **arXiv:2502.14529**.
- [45] B. McMahan, E. Moore, D. Ramage, S. Hampson, B. Agüera y Arcas, Communication-efficient learning of deep networks from decentralized data, in: Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 54, 2017, pp. 1273–1282. **arXiv:1602.05629**.
- [46] E. M. El Mhamdi, R. Guerraoui, S. Rouault, The hidden vulnerability of distributed learning in Byzantium, in: Proceedings of the 35th International Conference on Machine Learning (ICML), PMLR 80, 2018, pp. 3521–3530. **arXiv:1802.07927**.
- [47] H. P. Lopuhaä, P. J. Rousseeuw, Breakdown points of affine equivariant estimators of multivariate location and covariance matrices, *The Annals of Statistics* 19 (1) (1991) 229–248. doi:10.1214/aos/1176347978.
- [48] S. Minsker, Geometric median and robust estimation in Banach spaces, *Bernoulli* 21 (4) (2015) 2308–2335. doi:10.3150/14-BEJ645.
- [49] I. Diakonikolas, G. Kamath, D. M. Kane, J. Li, A. Moitra, A. Stewart, Robust estimators in high dimensions without the computational intractability, in: 2016 IEEE 57th Annual Symposium on Foundations of Computer Science (FOCS), 2016, pp. 655–664. **arXiv:1604.06443**.
- [50] K. A. Lai, A. B. Rao, S. Vempala, Agnostic estimation of mean and covariance, in: 2016 IEEE 57th Annual Symposium on Foundations of Computer Science (FOCS), 2016, pp. 665–674. doi:10.1109/FOCS.2016.76.
- [51] I. Diakonikolas, D. M. Kane, Recent advances in algorithmic high-dimensional robust statistics, arXiv preprint arXiv:1911.05911 (2019). **arXiv:1911.05911**.
- [52] E. Bagdasaryan, A. Veit, Y. Hua, D. Estrin, V. Shmatikov, How to backdoor federated learning, in: Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 108, 2020, pp. 2938–2948. **arXiv:1807.00459**.

- [53] L. Lamport, R. Shostak, M. Pease, The Byzantine generals problem, *ACM Transactions on Programming Languages and Systems (TOPLAS)* 4 (3) (1982) 382–401. doi:10.1145/357172.357176.
- [54] M. Castro, B. Liskov, Practical Byzantine fault tolerance, in: *Proceedings of the Third Symposium on Operating Systems Design and Implementation (OSDI)*, 1999, pp. 173–186.
- [55] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017. arXiv:1706.04599.
- [56] N. Reimers, I. Gurevych, Sentence-BERT: Sentence embeddings using Siamese BERT-networks, in: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019. arXiv:1908.10084.
- [57] T. Gao, X. Yao, D. Chen, SimCSE: Simple contrastive learning of sentence embeddings, in: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021. arXiv:2104.08821.
- [58] J. Bernstein, J. Zhao, K. Azizzadenesheli, A. Anandkumar, signSGD with majority vote is communication efficient and fault tolerant, in: *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1810.05291.
- [59] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Aspell, P. Welinder, P. Christiano, J. Leike, R. Lowe, Training language models to follow instructions with human feedback, in: *Advances in Neural Information Processing Systems (NeurIPS)* 35, 2022. arXiv:2203.02155.
- [60] Y. Bai, S. Kadavath, S. Kundu, A. Aspell, J. Kernion, A. Jones, A. Chen, A. Goldie, A. Mirhoseini, C. McKinnon, et al., Constitutional AI: Harmlessness from AI feedback, arXiv preprint arXiv:2212.08073 (2022). arXiv:2212.08073.
- [61] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, S. Batra, P. Bhargava, S. Bhosale, et al., Llama 2: Open

- foundation and fine-tuned chat models, arXiv preprint arXiv:2307.09288 (2023). [arXiv:2307.09288](#).
- [62] A. Mehrotra, M. Zampetakis, P. Kassianik, B. Nelson, H. Anderson, Y. Singer, A. Karbasi, Tree of attacks: Jailbreaking black-box LLMs automatically, in: Advances in Neural Information Processing Systems (NeurIPS) 37, 2024. [arXiv:2312.02119](#).
 - [63] J. Yu, X. Lin, Z. Yu, X. Xing, GPTFUZZER: Red teaming large language models with auto-generated jailbreak prompts, arXiv preprint arXiv:2309.10253 (2023). [arXiv:2309.10253](#).
 - [64] M. Mazeika, L. Phan, X. Yin, A. Zou, Z. Wang, N. Mu, E. Sakhaee, N. Li, S. Basart, B. Li, D. Forsyth, D. Hendrycks, HarmBench: A standardized evaluation framework for automated red teaming and robust refusal, in: Proceedings of the 41st International Conference on Machine Learning (ICML), 2024. [arXiv:2402.04249](#).
 - [65] Q. Zhan, Z. Liang, Z. Ying, D. Kang, InjecAgent: Benchmarking indirect prompt injections in tool-integrated large language model agents, in: Findings of the Association for Computational Linguistics (ACL Findings), 2024. [arXiv:2403.02691](#).
 - [66] X. Liu, Z. Yu, Y. Zhang, N. Zhang, C. Xiao, Automatic and universal prompt injection attacks against large language models, arXiv preprint arXiv:2403.04957 (2024). [arXiv:2403.04957](#).